Uniform Laguerre signs and hypercomplex jets of the theta heat flow

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Abstract

For the theta heat family H, the two forms $H_x^2 - HH_{xx}$ and $H_{xx}^2 - H_xH_{xxx}$ control the evolution of its first jet at genuine stationary points. We prove both forms strictly positive, uniformly for $0 < t \le 1/2$ and sufficiently large $|x|$, when $t\log(|x|/(4\pi)) \ge 3 + \varepsilon$ for any fixed $\varepsilon > 0$. Oscillatory arithmetic blocks and Polymath's effective Riemann–Siegel approximation give a common nonzero amplitude for all four derivatives; shrinking complex discs pay the derivative errors. A separate explicit bound applies at $|x| \ge 4\pi\exp(10^9)$ and $t\log(|x|/(4\pi)) \ge 5$. Quaternionic and octonionic fields express the signed observations geometrically. Exact paired theta integrals retain all arithmetic and integration cross terms and give complete omitted-pair payments. Complete-source certificates establish both stationary signs and a physical first-jet gap through $|x| \le 104$. Prescribed source-filter pilots demonstrate direct absorption of an adverse residual, while endpoint and Fourier arguments identify limitations of fixed compact filters. Full-source Gaussian filters give exact cross-time identities and a further finite signed comparison. The all-height stationary bound near possible collisions and the finite transition strip remain open.

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1 The theta heat flow and the main result

The four real derivatives

$$h_j(t,x) = \partial_x^j H(t,x), \qquad j = 0,1,2,3, \tag{1.1}$$

form the smallest consecutive jet containing the collision detector (H, H_x) and both quadratic expressions

$$\mathcal{L}_0 = h_1^2 - h_0 h_2, \qquad \mathcal{L}_1 = h_2^2 - h_1 h_3. \tag{1.2}$$

Here "four jets" means these four consecutive derivative components. Their common scalar origin is essential: they are not four independent fields. The first two give the quaternionic observation

$F = h_0 + \mathbf{j}h_1$ and its energy $E = h_0^2 + h_1^2$. Since $H_t = -H_{xx}$,

$$\frac{1}{2}E_t = -h_0h_2 - h_1h_3 = \begin{cases} \mathcal{L}_0, & h_1 = 0, \\ \mathcal{L}_1, & h_2 = 0. \end{cases} \tag{1.3}$$

The stationary sets in this identity are those of the original scalar function and its first derivative. Their signs will remain the same objects throughout the analytic and hypercomplex constructions.

1.1 The fixed source and normalization

The family under consideration is

$$\Phi(u) = \sum_{n\geq 1} \phi_n(u), \qquad \phi_n(u) = (2\pi^2 n^4 e^{9u} - 3\pi n^2 e^{5u}) e^{-\pi n^2 e^{4u}}, \tag{1.4}$$

$$H(t,z) = \int_0^\infty e^{tu^2} \Phi(u) \cos(zu)\, du, \qquad H(0,z) = \frac{1}{8}\xi\left(\frac{1}{2} + \frac{iz}{2}\right). \tag{1.5}$$

We use the convention $\xi(s) = s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)/2$, agreeing with [3]. In particular the critical line corresponds to real z, and the heat equation has the backward sign

$$\partial_t H = -\partial_z^2 H. \tag{1.6}$$

For completeness, the normalization follows directly from the theta Mellin identity. Put $A(u) = e^u \sum_{n\geq 1} e^{-\pi n^2 e^{4u}}$. Poisson summation makes $P(u) = e^u + 2A(u)$ even, and differentiation gives $\Phi = (P'' - P)/16 = (A'' - A)/8$. Splitting the theta Mellin integral at one and substituting $y = e^{4u}$ gives

$$\xi(1/2 + iz/2) = \frac{1}{2} - (1+z^2)\int_0^\infty A(u)\cos(zu)\, du.$$

Evenness of P gives $A'(0) = -1/2$. Two integrations by parts now prove (1.5); all positive-infinity boundary terms vanish. This also proves that the complete source Φ is even. For $u \geq 0$ its terms are positive, and their successive majorant ratios yield

$$0 < \Phi(u) < 4\pi^2 e^{9u - \pi e^{4u}}. \tag{1.7}$$

Indeed the ratio of successive terms of $n^4 e^{-\pi n^2 e^{4u}}$ is at most $16e^{-3\pi} < 1/2$. The bound justifies differentiation of every order on compact sets of complex (t,z). Thus H is jointly entire, even in z, and real on real (t,z); (1.6) follows under the integral. For each fixed real time it is a real entire function of order at most one, as is also immediate from the same source bound by splitting the integral at $u = O(\log(2 + |z|))$.

1.2 Uniform positivity outside the inner region

The leading result concerns the dimensionless parameter $t \log|x|$, which remains nontrivial as time tends to zero while height increases.

Theorem 1.1 (Uniform four-jet asymptotic)**.** *For every $\varepsilon > 0$ there is $X_\varepsilon > 4\pi$ with the following property. Uniformly for*

$$x \geq X_\varepsilon, \qquad 0 < t \leq \tfrac{1}{2}, \qquad \theta := tL \geq 3 + \varepsilon, \qquad L := \log\frac{x}{4\pi}, \tag{1.8}$$

there is an explicitly defined nonzero complex amplitude $Q = Q_t(x)$ such that, with $\omega = L/4$ and $\phi = \arg Q$,

$$\frac{1}{2|Q|}\left(h_0, \frac{h_1}{\omega}, \frac{h_2}{\omega^2}, \frac{h_3}{\omega^3}\right) = (\cos\phi, -\sin\phi, -\cos\phi, \sin\phi) + O_\varepsilon(L^{-1}). \tag{1.9}$$

Consequently

$$h_0^2 + (h_1/\omega)^2 = 4|Q|^2(1 + O_\varepsilon(L^{-1})), \tag{1.10}$$

$$\mathcal{L}_0 = 4|Q|^2\omega^2(1 + O_\varepsilon(L^{-1})), \qquad \mathcal{L}_1 = 4|Q|^2\omega^4(1 + O_\varepsilon(L^{-1})). \tag{1.11}$$

After increasing X_ε, both Laguerre forms are strictly positive everywhere in (1.8), and $F(t,x) \neq 0$ there. Evenness gives the reflected statements at negative heights.

The theorem combines the effective approximation in [3, Theorem 1.3] with oscillatory arithmetic blocks, uniformly in all four derivatives. The cutoff for $3 + \varepsilon$ has not been evaluated numerically. Three is a sufficient threshold of this argument; no optimality or boundary-case claim at $\varepsilon = 0$ is made. A different, more conservative region has a fully explicit height.

Proposition 1.2 (An explicit outer region)**.** *Set $X_{\exp} = 4\pi\exp(10^9)$. For $x \geq X_{\exp}$, $0 < t \leq 1/2$, and $tL \geq 5$, every coordinate error in (1.9) is at most $108000000/L \leq 0.108 < 1/8$. Consequently*

$$\frac{h_0^2 + (h_1/\omega)^2}{4|Q|^2} \geq \frac{1}{2}, \qquad \frac{\mathcal{L}_j}{4|Q|^2\omega^{2j+2}} \geq \frac{31}{64}, \quad j = 0, 1. \tag{1.12}$$

Reflection gives the negative-height statements.

The complete explicit estimates are in Appendix B. This enormous cutoff does not provide a practical bridge from height 104, and it is not an evaluated cutoff for the sharper $3 + \varepsilon$ theorem.

The geometric interpretation is that a common nonzero amplitude controls all four jets, including at phases where individual leading components vanish. In quaternionic and octonionic coordinates, the same quadratic forms become signed geometric observations. Those constructions do not supply positivity by themselves; they identify precisely which part of the theta analysis is needed. A second, complementary result is the validated bounded cover

$$|F(t,x)| > 10^{-16} \quad (0 \leq t \leq 1/2,\ |x| \leq 104), \tag{1.13}$$

together with both strict stationary signs there. Its complete-source proof and certificate structure are given in Section 6.

Real double zeros at positive times are the relevant obstruction: the classical heat-flow inputs imply

$$\text{RH} \iff F(t,x) \neq 0 \quad \text{for every } t > 0 \text{ and } x \in \mathbb{R}. \tag{1.14}$$

Section 3 gives the reduction and the sufficient stationary bounded-loss criterion. Theorem 1.1 and (1.13) leave an inner small-time region and a finite transition strip. The exact shared-source formula later in the paper formulates the remaining arithmetic estimate without discarding its signed cross terms.

2 Proof of the uniform outer-region theorem

2.1 A holomorphic comparison amplitude

Use the principal logarithm and set

$$M_0(s) = \frac{s(s-1)}{16}\pi^{-s/2}\sqrt{2\pi}\exp\left\{\left(\frac{s}{2} - \frac{1}{2}\right)\operatorname{Log}\frac{s}{2} - \frac{s}{2}\right\}, \tag{2.1}$$

$$\alpha(s)=\frac{M_0'(s)}{M_0(s)}=\frac{1}{2s}+\frac{1}{s-1}+\frac{1}{2}\operatorname{Log}\frac{s}{2\pi},\qquad M_t(s)=e^{t\alpha(s)^2/4}M_0(s), \tag{2.2}$$

$$s_\pm(z)=\frac{1\pm iz}{2},\qquad w_\pm(z)=s_\pm(z)+\frac{t}{2}\alpha(s_\pm(z)), \tag{2.3}$$

$$Q_\pm(z)=M_t(s_\pm(z))\zeta(w_\pm(z)),\qquad \mathcal{P}_t(z)=Q_+(z)+Q_-(z). \tag{2.4}$$

For real x we take $Q=Q_+(x)$; the other branch is its conjugate. The functions in these definitions are analytic on the small discs used below. Their arguments stay at imaginary parts approximately $\pm x/2$, away from all logarithmic cuts, while $\Re w_\pm>1$ keeps the zeta arguments away from the pole.

Write $z=u+iv$, $L_u=\log(u/(4\pi))$, and

$$N=\left\lfloor\sqrt{u/(4\pi)+t/16}\right\rfloor,\qquad S_{t,N}(w)=\sum_{n\le N}e^{t\log^2 n/4}n^{-w}.$$

Polymath's estimate applies when $u\ge 200$, $0\le v\le 1$, and $0<t\le 1/2$. In the present notation it reads

$$H(t,z)=M_t(s_-(z))S_{t,N}(w_-(z))+M_t(s_+(z))S_{t,N}(w_+(z))+\mathcal{E}(t,z), \tag{2.5}$$

where $|\mathcal{E}|\le|M_t(s_-)|(e_A+e_B+e_{C,0})$. This is the exact two-branch expression of the cited theorem: its first exponent s_* equals w_- and its second exponent, after incorporating the factor n^v, is $\overline{s_*}+\kappa-v=w_+$. To retain the full imported error information, define

$$\gamma=\frac{M_t(s_+)}{M_t(s_-)},\qquad \kappa=\frac{t}{2}\left(\alpha\left(\frac{1-v+iu}{2}\right)-\alpha\left(\frac{1+v+iu}{2}\right)\right).$$

The bounds needed from [3, eqs. (20)–(24)] are

$$|\gamma|\le e^{0.02v}(u/(4\pi))^{-v/2},\qquad |\kappa|\le\frac{tv}{2(u-6)}, \tag{2.6}$$

$$e_A+e_B\le\sum_{n\le N}(1+|\gamma|N^{|\kappa|}n^v)e^{t\log^2 n/4}n^{-\Re w_-}\left[\exp\left\{\frac{(t^2/16)\log^2(u/(4\pi n^2))+0.626}{u-6.66}\right\}-1\right], \tag{2.7}$$

$$e_{C,0}\le(u/(4\pi))^{-(1+v)/4}\exp\left\{-\frac{tL_u^2}{16}+\frac{1.24(3^v+3^{-v})}{N-0.125}+\frac{3|L_u+i\pi/2|+10.44}{u-12}\right\}. \tag{2.8}$$

No derivatives of this finite approximation will be taken. Its cutoff can jump with u; the analytic model (2.4) will remove that difficulty before differentiation.

2.2 A maximal estimate for logarithmic-phase blocks

The gain over absolute summation comes from retaining the oscillation in n^{-iT} before estimating each arithmetic block.

Lemma 2.1 (Logarithmic-phase blocks)**.** *For $T\ge 1$, $M\ge 1$, and every interval I of integers contained in $(M,2M]$,*

$$\left|\sum_{n\in I}n^{-iT}\right|\ll T^{1/6}M^{1/2}+T^{-1/6}M. \tag{2.9}$$

The implied constant is absolute. The same estimate holds with T replaced by $-T$, and is uniform over all shortened terminal blocks.

Proof. Apply the third derivative test, in the form of [4, Theorem 2 and its preceding proof], to $f(y) = -(T/(2\pi)) \log y$. On $(M, 2M]$ its third derivative has magnitude comparable to T/M^3, with an absolute comparison ratio. For an interval of length $J \le M$, the test gives

$$J(T/M^3)^{1/6} + J^{1/2}(T/M^3)^{-1/6} \le T^{1/6}M^{1/2} + T^{-1/6}M.$$

Empty intervals contribute zero; one-point intervals are absorbed by increasing the absolute constant. Conjugation handles the opposite phase. □

For a continuously differentiable, possibly complex weight W, Abel summation on a shortened block is the exact identity

$$\sum_{M<n\le R} W(n)n^{-iT} = W(R)S_M(R) - \int_M^R W'(y)S_M(y)\,dy, \quad S_M(y) = \sum_{M<n\le y} n^{-iT}, \quad M \le R \le 2M. \tag{2.10}$$

Thus (2.9) is multiplied by $|W(R)| + \int_M^R |W'(y)|\,dy$. For a nonnegative decreasing weight, this total-variation factor is exactly $W(M)$. The maximal form of the lemma is needed because the last block ends at the actual cutoff N.

2.3 Uniform replacement of the weighted sums above three

Fix (t, x) in (1.8). It is enough to consider $0 < \varepsilon \le 1$, since larger positive gaps give smaller domains. Set

$$D_x = \{z : |z - x| \le L^{-1}\}, \qquad L_u = \log(u/(4\pi)), \qquad \theta_u = tL_u. \tag{2.11}$$

Throughout this disc, $L_u = L + O((xL)^{-1})$ and $|v| \le L^{-1}$. For sufficiently large x depending on ε, the elementary formula (2.2) gives, for either branch $w = \sigma + i\tau = w_\pm(z)$,

$$\begin{aligned} &\theta_u \ge 3 + \varepsilon/2, \qquad |\tau| \asymp u, \qquad \log N \le L_u/2 + O(u^{-1}), \\ &\sigma \ge \tfrac{1}{2} + \tfrac{\theta_u}{4} - \delta_x, \qquad 0 \le \delta_x = O(L^{-1}), \qquad \delta_x \le \varepsilon/72. \end{aligned} \tag{2.12}$$

The constants are uniform for $0 < t \le 1/2$, including when tL grows. In particular, $\sigma \ge 5/4$. The full weighted absolute majorant still has the form

$$e^{t\log^2 n/4}n^{-\sigma} \le n^{-p}, \qquad p = \frac{1}{2} + \frac{\theta_u}{8} - O(L^{-1}), \qquad n \le N, \tag{2.13}$$

but it need not be summable when $3 < \theta_u \le 4$. We use absolute summation only for the smaller indices and oscillatory blocks for the rest.

Lemma 2.2 (Weighted logarithmic moments)**.** *Under* (2.12)*, for every fixed nonnegative integer r,*

$$\sum_{n\le N} (\log n)^r e^{t\log^2 n/4}n^{-w} - (-1)^r\zeta^{(r)}(w) = O_{\varepsilon,r}(L^{-1}), \qquad z \in D_x. \tag{2.14}$$

For $r = 0$ the factor at $n = 1$ is interpreted as one. In particular the estimate is simultaneous for $r = 0, 1, 2, 3$.

Proof. Write $\ell = L_u$ and split at the real number $Y = e^{\ell/3}$, with $n \le Y$ interpreted as the exact integer cutoff. For sufficiently large x one has $Y < N$. If $n \le Y$, the inequality $e^q - 1 \le qe^q$ gives

$$\left|\sum_{n\le Y} (\log n)^r (e^{t\log^2 n/4} - 1)n^{-w}\right| \le \frac{t}{4}\sum_{n=2}^{\infty} (\log n)^{r+2}n^{-(1/2+\theta_u/6-\delta_x)}. \tag{2.15}$$

Put $\theta_0 = 3 + \varepsilon/2$ and $s = \theta_u - \theta_0 \geq 0$. The exponent on the right satisfies

$$\frac{1}{2} + \frac{\theta_u}{6} - \delta_x \geq 1 + \frac{\varepsilon}{24} + \frac{s}{6}. \tag{2.16}$$

Consequently, with $D_{\varepsilon,r} = \sum_{n\geq 2} (\log n)^{r+2} n^{-1-\varepsilon/24} < \infty$, its right side is at most

$$\frac{D_{\varepsilon,r}}{4\ell}(\theta_0 + s)2^{-s/6} = O_{\varepsilon,r}(\ell^{-1}). \tag{2.17}$$

The boundedness of the numerator for every $s \geq 0$ supplies the required uniformity in time. The unweighted zeta tail beyond Y is bounded by

$$\sum_{n>Y} (\log n)^r n^{-\sigma} \ll_r (1 + \log^r Y) Y^{-1/4} \ll_r \ell^r e^{-\ell/12}, \tag{2.18}$$

using $\sigma \geq 5/4$.

It remains to estimate the weighted terms $Y < n \leq N$. Partition them into the exact intervals $(M, \min(2M, N)]$, starting at $M = Y$ and successively doubling. On each block put

$$W_r(y) = (\log y)^r e^{t \log^2 y/4} y^{-\sigma}.$$

This weight is nonnegative and, for sufficiently large ℓ, monotonically decreasing: throughout $Y \leq y \leq N$,

$$y(\log W_r)' = \frac{r}{\log y} + \frac{t}{2}\log y - \sigma \leq \frac{3r}{\ell} - \frac{1}{2} + \delta_x + O(u^{-1}) < 0. \tag{2.19}$$

Apply Lemma 2.1 with $T = |\tau| \asymp u$ and (2.10). The block has modulus at most

$$\ll W_r(M)\{u^{1/6} M^{1/2} + u^{-1/6} M\}. \tag{2.20}$$

Let $\beta = (\log M)/\ell$. Apart from a factor $O_r(\ell^r)$ and bounded constants, the two terms in (2.20) are bounded by $e^{E_1 \ell}$ and $e^{E_2 \ell}$, where

$$\begin{aligned} E_1 &= \frac{1}{6} - \frac{\theta_u}{4}\beta(1-\beta) + \delta_x \beta, \\ E_2 &= -\frac{1}{6} + \frac{\beta}{2} - \frac{\theta_u}{4}\beta(1-\beta) + \delta_x \beta. \end{aligned} \tag{2.21}$$

On $1/3 \leq \beta \leq 1/2$, one has $\beta(1-\beta) \geq 2/9$, hence

$$E_1 \leq \frac{1}{6} - \frac{\theta_u}{18} + \frac{\delta_x}{2} \leq -\frac{\varepsilon}{48}.$$

For the second exponent at $\theta_u = 3$, its convex quadratic part has endpoint values $-1/6$ and $-5/48$; increasing θ_u decreases it. Together with $\delta_x \leq \varepsilon/72 \leq 1/72$, this gives $E_2 \leq -1/16$ after an arbitrarily small increase of the lower height. The possible extension to $\beta \leq 1/2 + O((u\ell)^{-1})$ changes these exponents by $O(u^{-1})$ uniformly, since $\theta_u \leq \ell/2$. It can therefore be absorbed by weakening the first exponential decay to $\varepsilon/100$. There are $O(\ell)$ blocks, so, with $c_\varepsilon = \min(\varepsilon/100, 1/16) > 0$,

$$\sum_{Y<n\leq N} (\log n)^r e^{t \log^2 n/4} n^{-w} = O_{\varepsilon,r}(\ell^{r+1} e^{-c_\varepsilon \ell}). \tag{2.22}$$

Combining (2.15), (2.18), and (2.22) proves the claim. The final block is shortened at the actual N; no derivative of that cutoff has been taken. □

2.4 Complete value errors and derivative control

The positive weight sums in the imported errors do not require the summability of (2.13). For $n \le N$,

$$e^{t\log^2 n/4} n^{-\theta_u/4} = \exp\{-\tfrac{t}{4}\log n(L_u - \log n)\} \ll 1, \qquad n^{\delta_x} \ll 1.$$

The second estimate uses $\delta_x = O(L^{-1})$ and $\log N = O(L)$. On the upper half-disc, (2.6) also bounds $|\gamma|$ and $N^{|\kappa|}$ uniformly. Multiplication by n^v replaces the first branch's real part $(1+v)/2$ by $(1-v)/2$, already covered by (2.12). Hence the entire positive weight sum in (2.7) is $O(\sum_{n\le N} n^{-1/2}) = O(u^{1/4})$. The per-index error multiplier is $O(L^2/u)$, uniformly for $t \le 1/2$, because $|\log(u/(4\pi n^2))| \le L_u + O(u^{-1})$. Thus all three imported errors are paid by

$$e_A + e_B = O(L^2 u^{-3/4}), \qquad e_{C,0} \ll e^{-L_u/4 - \theta_u L_u/16} \ll u^{-7/16}. \tag{2.23}$$

For the second estimate the positive correction exponent in (2.8) tends uniformly to zero.

Let $\Gamma = |M_t(s_+(x))|$. The logarithmic z-derivative of either $M_t(s_\pm(z))$ is $O(L)$ uniformly for $0 < t \le 1/2$. Integration along a segment of length at most $1/L$ gives

$$|M_t(s_\pm(z))| \asymp \Gamma \qquad (z \in D_x). \tag{2.24}$$

This is why the complex disc shrinks: a fixed-radius disc would permit powers of x in the amplitude comparison. Combining (2.5), Lemma 2.2 at $r = 0$, and (2.23)–(2.24), we obtain

$$H(t,z) - \mathcal{P}_t(z) = O_\varepsilon(\Gamma/L). \tag{2.25}$$

Initially the imported approximation proves this on the upper half-disc. Schwarz symmetry of H and $\mathcal{P}_t$ extends it to the lower half.

Absolute convergence of the Dirichlet series for ζ and $1/\zeta$ in $\Re w \ge 5/4$ gives

$$\zeta(5/4)^{-1} \le |\zeta(w)| \le \zeta(5/4). \tag{2.26}$$

Thus $Q \ne 0$ and $|Q| \asymp \Gamma$, using only a known zero-free half-plane. The difference in (2.25) is holomorphic on a neighborhood of the closed disc. Cauchy's formula, with time fixed, therefore yields

$$\partial_x^k (H(t,\cdot) - \mathcal{P}_t)(x) = O_{\varepsilon,k}(|Q| L^{k-1}), \qquad k = 0,1,2,3. \tag{2.27}$$

At $k = 0$ this means $O_\varepsilon(|Q|/L)$. Only the holomorphic difference is differentiated; the discontinuous arithmetic cutoff is absent from that step.

2.5 The common phase and the two signs

At the real center,

$$\alpha(s_+(x)) = L/2 + i\pi/4 + O(x^{-1}), \qquad \alpha^{(r)}(s_+(x)) = O_r(x^{-r}) \quad (r \ge 1). \tag{2.28}$$

The absolutely convergent prime-power series bounds every fixed logarithmic derivative of ζ uniformly in $\Re w \ge 5/4$. Differentiating a local logarithm of Q_+ gives

$$(\log Q_+)' = \frac{i}{2}\left[\alpha + \frac{t}{2}\alpha\alpha' + \left(1 + \frac{t}{2}\alpha'\right)\frac{\zeta'(w_+)}{\zeta(w_+)}\right] = i\omega + O(1), \tag{2.29}$$

$$(\log Q_+)'' = O(1), \qquad (\log Q_+)''' = O(1). \tag{2.30}$$

All α terms are evaluated at $s_+(x)$, and time is fixed. The bounds remain uniform as $\theta = tL$ grows because $t \le 1/2$. The exponential derivative formulas imply

$$Q_+^{(k)}(x) = Q\big((i\omega)^k + O(L^{k-1})\big), \qquad k = 1, 2, 3.$$

The real-axis derivatives of Q_- are their conjugates. Together with (2.27), this proves (1.9).

Set $P_k = h_k/(2|Q|\omega^k)$ and write $P = (c, -s, -c, s) + e$, where $c^2 + s^2 = 1$ and each component of e is $O_\varepsilon(L^{-1})$. Direct expansion gives

$$P_0^2 + P_1^2 = 1 + O_\varepsilon(L^{-1}), \quad P_1^2 - P_0P_2 = 1 + O_\varepsilon(L^{-1}), \quad P_2^2 - P_1P_3 = 1 + O_\varepsilon(L^{-1}).$$

This proves (1.10)–(1.11), uniformly in phase, and completes the proof of Theorem 1.1. More explicitly, if $|e_j| \le d$, then

$$P_0^2 + P_1^2 \ge 1 - 4d, \qquad P_1^2 - P_0P_2,\ P_2^2 - P_1P_3 \ge 1 - 4d - d^2. \tag{2.31}$$

For example the first signed form is $1 - 2se_1 + e_1^2 - ce_2 + ce_0 - e_0e_2$; retain its nonnegative square and bound the remaining terms. At $d = 1/8$, the two signed margins are at least $31/64$ and the balanced norm is at least $1/2$. Appendix B gives a fully explicit height at the stronger condition $tL \ge 5$. The threshold-three theorem has an unevaluated height and asserts no optimality of the number three.

Corollary 2.3 (Asymptotic collision localization)**.** *If $H(T_m, a_m) = H_x(T_m, a_m) = 0$, $0 < T_m \le 1/2$, and $|a_m| \to \infty$, then*

$$\limsup_{m\to\infty} T_m \log \frac{|a_m|}{4\pi} \le 3.$$

Proof. For each fixed $\varepsilon > 0$, Theorem 1.1 excludes every sufficiently high term whose displayed product is at least $3 + \varepsilon$. □

This is an asymptotic localization. It does not impose the constant three at every finite height, and setting $\varepsilon = 0$ in the uniform estimates is not justified.

3 Why positive-time collisions are the obstruction

The preceding asymptotic result is useful because a failure of RH forces a finite collision of real zeros at a positive time. We record the precise reduction, including the compactness input, before formulating the signed estimate still needed outside the proved regions.

We use the normalization

$$H(0, z) = \frac{1}{8}\xi\left(\frac{1}{2} + \frac{iz}{2}\right), \qquad H_t = -H_{zz}.$$

The positive, rapidly decreasing theta source makes H jointly entire in (t, z), even in z, and real on the real (t, z) plane. Moreover, $H(t, iy) > 0$ for real t, y, so no time slice is identically zero and every zero has finite multiplicity.

The classical de Bruijn–Newman theory gives a finite real threshold Λ such that all zeros of $H(t, \cdot)$ are real exactly when $t \ge \Lambda$; the endpoint is included [1, 2]. In this normalization $\Lambda \le 1/2$ and RH is equivalent to $\Lambda \le 0$. De Bruijn's strip contraction keeps all zeros in $|\operatorname{Im} z| \le 1$ for $t \ge 0$. We also use the uniform confinement theorem of [3, Theorem 1.5(i)]: there is an absolute constant $C > 0$ such that, for $0 < t \le 1/2$, a nonreal zero satisfies

$$|\operatorname{Re} z| < e^{C/t}. \tag{3.1}$$

Uniformity in time, rather than a separate uncontrolled radius for each slice, is what prevents a first collision from escaping to infinity.

Lemma 3.1 (Local splitting at a multiple zero)**.** *Let W be real analytic near (T, a), solve $W_t = -W_{xx}$, and have an exact spatial zero of multiplicity $m \geq 2$ there. Its nearby zeros have leading locations*

$$a + \sqrt{2(t-T)}\,\lambda_k + O(|t-T|), \qquad k = 1, \dots, m,$$

where the λ_k are the distinct real roots of the probabilists' Hermite polynomial He_m*. In particular, W has a nonreal zero at times immediately before T.*

Proof. Put $W_q = \partial_x^q W(T, a)$. The convergent heat Taylor series is

$$W(T+\delta, a+w) = \sum_{p,q\geq 0} \frac{(-\delta)^p w^q}{p!q!} W_{2p+q}.$$

With $\delta = \eta^2$ and $w = \sqrt{2}\eta\lambda$, division by η^m extends analytically to $\eta = 0$, where its value is $2^{m/2} W_m \,\mathrm{He}_m(\lambda)/m!$. The roots of He_m are real and simple: its Rodrigues formula gives orthogonality to all lower-degree polynomials for the positive Gaussian weight, and the polynomial formed from its real sign-changing factors would contradict that orthogonality unless there were m such roots. The simple roots therefore continue by the implicit function theorem. For $t < T$, take $\eta = i\sqrt{T-t}$; at least one Hermite root is nonzero, so its leading imaginary displacement cannot be cancelled by the $O(T-t)$ remainder. □

Proposition 3.2 (Finite collision criterion for RH)**.** *RH holds if and only if*

$$H(t,x) = H_x(t,x) = 0 \quad \text{has no solution with } 0 < t \leq 1/2,\ x \in \mathbb{R}. \tag{3.2}$$

Equivalently, $E(t,x) = h_0(t,x)^2 + h_1(t,x)^2$ is positive throughout this region. No uniform positive lower bound for E is required. Equivalently, there is no such joint zero at any positive time.

Proof. If RH holds, all zeros are real at all nonnegative times. A multiple zero at a positive time would contradict Lemma 3.1.

Conversely, if RH fails then $0 < \Lambda \leq 1/2$. Choose $t_k \uparrow \Lambda$ with $t_k \geq \Lambda/2$, and a nonreal zero z_k at each t_k. The strip bound and (3.1) put these zeros in the fixed compact set $|\operatorname{Im} z| \leq 1$, $|\operatorname{Re} z| \leq e^{2C/\Lambda}$. A subsequence converges to a zero a of $H(\Lambda, \cdot)$, which is real by the threshold theorem. If that zero were simple, the local implicit zero branch would be unique; conjugation symmetry would make it real for real times on both sides of Λ. This contradicts $z_k \notin \mathbb{R}$. Thus $H(\Lambda, a) = H_x(\Lambda, a) = 0$. The source positivity gives $H(\Lambda, 0) > 0$, so $a \neq 0$. □

3.1 A sufficient estimate only on two stationary sets

Recall

$$\mathcal{L}_j = h_{j+1}^2 - h_j h_{j+2}, \qquad j = 0, 1.$$

On a smooth stationary curve $h_{j+1}(t, \chi(t)) = 0$, the heat equation gives

$$\frac{1}{2}\frac{d}{dt} h_j(t, \chi(t))^2 = \mathcal{L}_j(t, \chi(t)).$$

The following result includes singular stationary sets and all finite multiplicities. Its losses may grow with height or as $t \downarrow 0$; only local boundedness through each possible positive-time collision is required.

Lemma 3.3 (A stationary predecessor of an even zero)**.** *Suppose $W_t = -W_{xx}$ and W has exact spatial multiplicity $2r$ at (T, a), where $r \geq 1$. With $W_q = \partial_x^q W(T, a)$, there is a real analytic function $\chi(h)$ for small $h \geq 0$ such that*

$$W_x(T-h, \chi(h)) = 0, \qquad \chi(h) = a - \frac{W_{2r+1}}{rW_{2r}}h + O(h^2),$$
$$W(T-h, \chi(h)) = \frac{W_{2r}}{r!}h^r + O(h^{r+1}), \qquad W_{xx}(T-h, \chi(h)) = \frac{W_{2r}}{(r-1)!}h^{r-1} + O(h^r).$$

Proof. Use the heat Taylor series above with $\delta = -h$ and $w = hy$. The function $h^{-r}W_x(T-h, a+hy)$ extends analytically to $h = 0$, where it equals $W_{2r}y/(r-1)! + W_{2r+1}/r!$. Its y derivative is nonzero. The implicit function theorem gives $y(h)$, and substitution in the same series gives the two displayed asymptotics. □

Theorem 3.4 (Stationary bounded-loss criterion)**.** *Suppose there are nonnegative functions K_0, K_1, locally bounded on $(0, 1/2] \times \mathbb{R}$, for which*

$$\begin{aligned} h_1 = 0 \quad &\Longrightarrow \quad \mathcal{L}_0 \geq -K_0 E, \\ h_2 = 0 \quad &\Longrightarrow \quad \mathcal{L}_1 \geq -K_1 E. \end{aligned} \tag{3.3}$$

Then RH holds. Local boundedness here includes a neighborhood of every candidate joint zero, using a left neighborhood in time at $t = 1/2$.

Proof. Suppose H has a real multiple zero at (T, a) in this time range, of exact multiplicity $m \geq 2$. If $m = 2r$, apply Lemma 3.3 to $W = H$. On its predecessor branch $h_1 = 0$, $E = H^2 > 0$, and

$$-\frac{\mathcal{L}_0}{E} = \frac{H_{xx}}{H} = \frac{r}{h} + O(1), \qquad h = T - t \downarrow 0.$$

This contradicts the first inequality with locally bounded K_0. If $m = 2r + 1$, apply the lemma to $W = H_x$. Then $h_2 = 0$, $H_x = ch^r + O(h^{r+1})$, and $H_{xxx} = rch^{r-1} + O(h^r)$, with $c \neq 0$. The heat Taylor series also gives $H = O(h^{r+1})$ on this branch, because its displacement from a is $O(h)$. Consequently

$$E = H_x^2(1 + O(h^2)), \qquad -\frac{\mathcal{L}_1}{E} = \frac{r}{h} + O(1),$$

contradicting the second inequality. All divisions occur on proved nonzero predecessor branches, not at the supposed collision. Proposition 3.2 now applies. □

Both levels matter: the heat polynomial $H(t, x) = x^3 - 6(t-T)x$ has $\mathcal{L}_0 = 3x^4 + 36(t-T)^2 \geq 0$ everywhere, but has a triple zero at $(T, 0)$. Its second stationary loss diverges on approaching that collision from earlier times. Thus positivity of the first observation alone is insufficient for arbitrary multiplicity.

4 The four jets as quaternionic and octonionic fields

The four scalar jets enter the collision problem in overlapping pairs: H, H_x detect a multiple zero, while the two stationary signs involve (H, H_x, H_{xx}) and (H_x, H_{xx}, H_{xxx}). Quaternion multiplication keeps both the norms of these pairs and their signed determinants. A compatible octonionic extension retains these observations as oriented associators. We give the constructions explicitly so that their analytic content and their limitations are visible.

4.1 A quaternionic heat field with independent external time

Use Hamilton units $\mathbf{ij} = \mathbf{k}$, $\mathbf{i}^2 = \mathbf{j}^2 = \mathbf{k}^2 = -1$, and identify $\mathbb{C}$ with $\mathbb{C}_\mathbf{i} = \mathbb{R} + \mathbf{i}\mathbb{R}$. Complex coefficients are placed on the right of $\mathbf{j}$. Thus

$$(a + \mathbf{j}b)(c + \mathbf{j}d) = ac - \bar{b}d + \mathbf{j}(bc + \bar{a}d), \qquad \overline{a + \mathbf{j}b} = \bar{a} - \mathbf{j}b, \tag{4.1}$$

for $a, b, c, d \in \mathbb{C}_\mathbf{i}$, and $|a + \mathbf{j}b|^2 = |a|^2 + |b|^2$.

Write the independent quaternion coordinates as

$$q = x_0 + \mathbf{i}x_1 + \mathbf{j}x_2 + \mathbf{k}x_3 = z_0 + \mathbf{j}z_1, \qquad z_0 = x_0 + \mathbf{i}x_1, \quad z_1 = x_2 - \mathbf{i}x_3.$$

The unnormalized left Fueter operator and its conjugate are

$$D_q = \partial_{x_0} + \mathbf{i}\partial_{x_1} + \mathbf{j}\partial_{x_2} + \mathbf{k}\partial_{x_3}, \qquad \overline{D}_q = \partial_{x_0} - \mathbf{i}\partial_{x_1} - \mathbf{j}\partial_{x_2} - \mathbf{k}\partial_{x_3}.$$

A differentiable field is *left Fueter regular* if $D_q F = 0$. The convention using the factor $1/2$ has the same kernel; all units here multiply on the left. For the classical theory see [5].

Let $H(\tau, z)$ be the jointly entire theta heat family, satisfying $H_\tau = -H_{zz}$. With $t \in \mathbb{R}$ an independent external parameter, set

$$\tau = t + \overline{z_1} = t + x_2 + \mathbf{i}x_3.$$

$$G_m(t, q) = \partial_z^m H(\tau, z_0) + \mathbf{j}\partial_z^{m+1} H(\tau, z_0), \qquad F = G_0. \tag{4.2}$$

The coordinate z_1 supplies a complex heat-time displacement; both entries of G_m have the same spectral argument.

Proposition 4.1 (Regular lift and jet identities)**.** *Each G_m is globally real analytic and left Fueter regular in q. For every $m \geq 0$,*

$$D_q G_m = 0, \qquad \overline{D}_q G_m = 2G_{m+1}, \qquad \partial_t G_m = -G_{m+2}. \tag{4.3}$$

$$\begin{aligned} \partial_{x_0} G_m &= G_{m+1}, & \partial_{x_1} G_m &= G_{m+1}\mathbf{i}, \\ \partial_{x_2} G_m &= -G_{m+2}, & \partial_{x_3} G_m &= -G_{m+2}\mathbf{i}. \end{aligned} \tag{4.4}$$

Proof. For complex-valued U, V, direct use of (4.1) gives

$$\tfrac{1}{2} D_q(U + \mathbf{j}V) = U_{\bar{z}_0} - V_{z_1} + \mathbf{j}(V_{z_0} + U_{\bar{z}_1}).$$

The derivatives are componentwise Wirtinger derivatives. Taking $U = \partial_z^m H(\tau, z_0)$ and $V = \partial_z^{m+1} H(\tau, z_0)$, the first two terms vanish by holomorphy and the final pair cancel by the heat equation. Joint entireness proves global real analyticity. Differentiating the declared arguments proves (4.4) and the external time identity. Finally $D_q + \overline{D}_q = 2\partial_{x_0}$ proves the remaining assertion. □

On the physical slice $q = x \in \mathbb{R}$, write $h_m = h_m(t, x) = \partial_x^m H(t, x)$. The jets are real and

$$F = h_0 + \mathbf{j}h_1, \qquad F_x = G_1 = h_1 + \mathbf{j}h_2, \qquad F_{xx} = G_2 = h_2 + \mathbf{j}h_3 = -F_t. \tag{4.5}$$

Thus the four physical jets occur in three consecutive overlapping quaternion pairs. The basic observations are

$$E = |F|^2 = h_0^2 + h_1^2, \qquad \mathcal{L}_j = h_{j+1}^2 - h_j h_{j+2} \quad (j = 0, 1). \tag{4.6}$$

In particular $F(t,x) = 0$ means exactly that $H(t,\cdot)$ has a multiple real zero at x. The scalar positive-time collision criterion therefore translates to

$$\text{RH} \quad \Longleftrightarrow \quad F(t,x) \neq 0 \quad (t > 0,\ x \in \mathbb{R}).$$

This uses the scalar collision theorem for the theta family. It makes no assertion about zeros at nonreal quaternion arguments.

The signed products and the heat equation give, on this slice,

$$\overline{G_j} G_{j+1} = h_j h_{j+1} + h_{j+1} h_{j+2} - \mathbf{j}\mathcal{L}_j, \tag{4.7}$$

$$E_t = -2(h_0 h_2 + h_1 h_3), \qquad (\partial_t + \partial_x^2) E = 2(h_1^2 + h_2^2). \tag{4.8}$$

Consequently

$$h_1 = 0 \implies E_t = 2\mathcal{L}_0, \qquad h_2 = 0 \implies E_t = 2\mathcal{L}_1. \tag{4.9}$$

These formulas preserve the two different stationary conditions. The positive source term in the backward-heat identity (4.8) does not give a forward minimum principle.

4.2 An eight-dimensional monogenic extension

For a reader unfamiliar with octonions, it suffices here to regard $\mathbb{O}$ as two copies of $\mathbb{H}$, with a specified multiplication. Write $(a,b) = a + \ell b$, where $\ell = (0,1)$, and adopt the Cayley–Dickson convention

$$(a,b)(c,d) = (ac - d\bar{b}, \bar{a}d + cb), \qquad \overline{(a,b)} = (\bar{a}, -b). \tag{4.10}$$

Then $|(a,b)|^2 = |a|^2 + |b|^2$. This is the convention in [6, Section 2.2]. Octonionic multiplication is nonassociative; we keep parentheses whenever three octonionic factors occur. All component products on the right of (4.10) are associative quaternion products.

Let $r = y_0 + \mathbf{i}y_1 + \mathbf{j}y_2 + \mathbf{k}y_3$, so that $p = q + \ell r$ is the eight-dimensional argument. Put $e_0 = 1, e_1 = \mathbf{i}, e_2 = \mathbf{j}, e_3 = \mathbf{k}$, and define the left operator

$$\mathcal{D}_8 = \sum_{\mu=0}^{3} e_\mu \partial_{x_\mu} + \sum_{\mu=0}^{3} (\ell e_\mu) \partial_{y_\mu}. \tag{4.11}$$

We call a field *left monogenic* if $\mathcal{D}_8 \mathcal{O} = 0$. This fixes the operator convention explicitly; general octonionic Cauchy–Riemann systems are discussed in [7].

Proposition 4.2 (Explicit octonionic extension)**.** *For each $m \geq 0$, the field*

$$\mathcal{O}_m(t,q,r) = \Big(G_m(t,q) - \tfrac{1}{2} G_{m+3}(t,q)\bar{r},\, G_{m+2}(t,q) \Big) \tag{4.12}$$

is globally real analytic and satisfies

$$\mathcal{D}_8 \mathcal{O}_m = 0, \qquad \partial_{x_0} \mathcal{O}_m = \mathcal{O}_{m+1}, \qquad \partial_t \mathcal{O}_m = -\mathcal{O}_{m+2}, \qquad \mathcal{O}_m(t,q,0) = (G_m, G_{m+2}). \tag{4.13}$$

Proof. Applying (4.10) separately to each term of (4.11) gives the component formula

$$\mathcal{D}_8(A,B) = \left(D_q A - \sum_{\mu=0}^{3} B_{y_\mu} \bar{e}_\mu,\, \overline{D}_q B + \sum_{\mu=0}^{3} A_{y_\mu} e_\mu \right). \tag{4.14}$$

For (4.12), B is independent of r. Moreover $D_q(G_{m+3}\overline{r}) = (D_qG_{m+3})\overline{r} = 0$, since the component algebra is associative and r is held fixed. The first component of (4.14) is therefore zero. The second is

$$2G_{m+3} - \tfrac{1}{2}\sum_{\mu=0}^{3} G_{m+3}\overline{e}_\mu e_\mu = 2G_{m+3} - 2G_{m+3} = 0.$$

The derivative identities follow directly from (4.3), and the explicit formula proves global real analyticity. □

The correction is necessary: the extension (G_m, G_{m+2}) constant in r has defect $(0, 2G_{m+3})$. More generally, two left Fueter-regular fields $A(q), B(q)$ admit the extension $(A - \frac{1}{2}B_{x_0}\overline{r}, B)$, by the same calculation. Thus the construction imposes no new condition on the original pair. For $m = 0$, its value away from $r = 0$ involves G_3, hence h_4; it is not a closed evolution of only four jets.

On the physical trace $(q, r) = (x, 0)$,

$$\mathcal{O}_0 = (h_0 + \mathbf{j}h_1) + \ell(h_2 + \mathbf{j}h_3) \in \mathrm{span}_{\mathbb{R}}\{1, \mathbf{j}, \ell, \ell\mathbf{j}\}.$$

This span is an associative quaternion subalgebra, as can be checked from (4.10). Its total norm is

$$|\mathcal{O}_0|^2 = E + h_2^2 + h_3^2. \tag{4.15}$$

At a nondegenerate double zero, $E = 0$ but $h_2 \neq 0$; the total octonionic norm is then positive. A collision criterion must retain the distinguished first pair.

4.3 Oriented associators preserve the signed observations

The *associator* measures the failure of associativity:

$$[A, B, C] = (AB)C - A(BC), \qquad \mathbf{u} = \ell\mathbf{j}.$$

For embedded quaternions $a, b \in \mathbb{H}$, the pair multiplication gives

$$[a, b, \ell] = \ell(\overline{ab} - \overline{ba}) = -\ell(ab - ba). \tag{4.16}$$

The last equality holds because a quaternion commutator has zero real part. The untwisted physical jets lie in the commutative plane $\mathbb{R} + \mathbf{j}\mathbb{R}$, so their associator with ℓ vanishes. Fixed right multiplication by $\mathbf{i}$ and $\mathbf{k}$ produces the needed signed quantities while preserving quaternion left regularity.

Lemma 4.3 (Dot products and determinants as associators)**.** *For $a = A + \mathbf{j}B$, $b = C + \mathbf{j}D$ with real A, B, C, D,*

$$[a\mathbf{i}, b\mathbf{i}, \ell] = -2\mathbf{u}(AD - BC), \qquad [a\mathbf{i}, b\mathbf{k}, \ell] = 2\mathbf{u}(AC + BD). \tag{4.17}$$

Proof. One has $a\mathbf{i} = A\mathbf{i} - B\mathbf{k}$, $b\mathbf{i} = C\mathbf{i} - D\mathbf{k}$, and $b\mathbf{k} = D\mathbf{i} + C\mathbf{k}$. Their quaternion commutators are, respectively, $2\mathbf{j}(AD - BC)$ and $-2\mathbf{j}(AC + BD)$. Substitute these into (4.16). □

In particular define, on the physical slice,

$$\mathcal{A}_E = [F\mathbf{i}, F\mathbf{k}, \ell] = 2\mathbf{u}E, \tag{4.18}$$

$$\mathcal{A}_j = [G_j\mathbf{i}, G_{j+1}\mathbf{i}, \ell] = 2\mathbf{u}\mathcal{L}_j \quad (j = 0, 1), \tag{4.19}$$

$$\mathcal{A}_t = [F\mathbf{i}, F_t\mathbf{k}, \ell] = \mathbf{u}E_t. \tag{4.20}$$

These are immediate from Lemma 4.3 and (4.8). In contrast to (4.15), $\mathcal{A}_E = 0$ is equivalent to $F = 0$. Equation (4.9) becomes

$$h_1 = 0 \implies \mathcal{A}_t = \mathcal{A}_0, \qquad h_2 = 0 \implies \mathcal{A}_t = \mathcal{A}_1.$$

There is no order on $\mathbb{O}$ being invoked. Specify the oriented ray $\mathcal{C}_{\mathbf{u}} = \{s\mathbf{u} : s \geq 0\}$. The two stationary sign conditions are exactly

$$\mathcal{A}_t \in \mathcal{C}_{\mathbf{u}} \quad \text{on} \quad \{h_1 h_2 = 0\}. \tag{4.21}$$

Likewise the locally bounded loss condition is exactly

$$\mathcal{A}_t + K\mathcal{A}_E \in \mathcal{C}_{\mathbf{u}} \quad \text{on} \quad \{h_1 h_2 = 0\}, \qquad K \geq 0, \tag{4.22}$$

with K locally bounded through every candidate collision. Indeed the coefficient on this ray is $E_t + 2KE$. Thus the associators preserve both the collision detector and the required orientation. Their norms alone would replace the signed quantities by absolute values.

4.4 Why a source-specific estimate remains necessary

The constructions above apply to every jointly entire backward-heat family real on the physical slice. The following control shows that even strict positivity and analytic decay of its cosine-transform source do not supply the missing sign.

Proposition 4.4 (A positive entire source with a positive-time collision)**.** *There exist strictly positive, even, entire functions Φ_*, decaying superexponentially on the real axis, whose heat cosine transforms are jointly entire and have a nondegenerate real double zero at a positive time. Their quaternionic and octonionic lifts satisfy all the preceding identities.*

Proof. Set $\rho(v) = 2e^{-v^4}/\Gamma(1/4)$, so that $\int_{\mathbb{R}} \rho(v)\,dv = 1$, and $\rho_\epsilon(v) = \epsilon^{-1}\rho(v/\epsilon)$ for real $\epsilon > 0$. Use

$$\Phi_\epsilon(u) = 2\{\rho_\epsilon(u-1) + \rho_\epsilon(u+1)\} + \rho_\epsilon(u-2) + \rho_\epsilon(u+2).$$

This source is strictly positive, even, and entire, with quartic decay on the real axis. Its half-line heat cosine transform is

$$\begin{aligned}\widetilde{H}_\epsilon(t,z) = 2\int_{\mathbb{R}} \rho(v) e^{t(1+\epsilon v)^2} \cos\bigl(z(1+\epsilon v)\bigr)\,dv \\ + \int_{\mathbb{R}} \rho(v) e^{t(2+\epsilon v)^2} \cos\bigl(z(2+\epsilon v)\bigr)\,dv.\end{aligned}$$

Quartic decay supplies an integrable majorant on every compact set of complex (t, z, ϵ). Hence this formula extends jointly entirely through $\epsilon = 0$, obeys backward heat, and there becomes

$$\widetilde{H}_0(t,z) = 2e^t \cos z + e^{4t}\cos(2z).$$

At $T_0 = (\log 2)/3$, $x_0 = \pi$, it satisfies $\widetilde{H}_0 = (\widetilde{H}_0)_x = 0$ and $(\widetilde{H}_0)_{xx} = -6e^{T_0} \neq 0$. By the heat equation the Jacobian of $(\widetilde{H}_\epsilon, (\widetilde{H}_\epsilon)_x)$ with respect to (t, x), at this point and $\epsilon = 0$, has determinant $-((\widetilde{H}_0)_{xx})^2 \neq 0$. The real analytic implicit-function theorem therefore gives a nearby nondegenerate real double zero at a positive time for every sufficiently small $\epsilon > 0$. □

The failure is visible in the signed observation. At such a collision (T, a), put $c = H_{xx}(T, a) \neq 0$. The implicit-function theorem gives a real stationary branch $H_x(t, \chi(t)) = 0$ through it. For $\delta = T - t \downarrow 0$, backward heat gives

$$H(t, \chi(t)) = c\delta + O(\delta^2), \qquad E(t, \chi(t)) = c^2\delta^2 + O(\delta^3), \qquad \mathcal{L}_0(t, \chi(t)) = -c^2\delta + O(\delta^2).$$

Thus (4.22) would require $K \geq \delta^{-1} + O(1)$, contradicting local boundedness at the collision. The lift and associator identities alone cannot provide that bound. For the theta family, its particular arithmetic structure must enter a further signed estimate.

For the genuine theta family, Theorem 1.1 supplies the additional sign information in its declared outer region: $\mathcal{A}_0$ and $\mathcal{A}_1$ lie strictly on $\mathcal{C}_{\mathbf{u}}$ there. At genuine stationary points the same is true of $\mathcal{A}_t$. This is the geometric consequence of the common arithmetic amplitude, not a consequence of monogenicity alone.

5 Centered residuals and the common theta carrier

The outer theorem uses oscillatory cancellation to reach a nonzero zeta comparison amplitude. To study the remaining region, we return to the exact theta source. The aim is to retain the same four derivatives and their signed correlations before estimating any error.

5.1 A contour identity for the physical derivatives

Proposition 5.1 (Complete-source contour shift)**.** *For real $t \geq 0$, $x \geq 0$, $0 < a < \pi/8$, and every integer $k \geq 0$,*

$$h_k(t, x) = e^{-ax}\Re \int_0^\infty (i(u + ia))^k e^{t(u+ia)^2} \Phi(u + ia) e^{ixu}\, du. \tag{5.1}$$

Proof. The theta series is normally convergent in the strip $|\Im z| < \pi/8$ because $\Re(e^{4z}) > 0$ there. Analytic continuation of the real reflection identity makes Φ even throughout the strip; Schwarz symmetry then makes $\Phi(iy)$ real. On the left edge of the contour shift, every factor of $(iz)^k e^{tz^2}\Phi(z)e^{ixz}$ is real at $z = iy$, while $dz = i\,dy$. Its contribution has zero real part. On the right edge, $\cos(4y) \geq \cos(4a) > 0$ gives uniform double-exponential decay. Cauchy's theorem proves the formula. The evenness argument applies to the complete source; it must precede any separation into individual theta summands. □

The all-index and integration bounds in Appendix A make truncation of (5.1) rigorous. They control absolute errors at the gamma exponential scale. Obtaining a sign still requires information about the retained, correlated jets.

5.2 A residual adapted to genuine stationarity

Freeze real a, b with $0 < a < \pi/8$ and $b \geq 0$, and put $\mu = a^2 + b^2$. For every integer $j \geq 0$, define

$$C_j = e^{-ax}\Re \int_0^\infty (u^2 - b^2)(i(u + ia))^j e^{t(u+ia)^2} \Phi(u + ia) e^{ixu}\, du. \tag{5.2}$$

The required stationary forms use $j = 0, 1$; higher residuals will enter exact heat-transport identities. The polynomial identity $u^2 - b^2 = -(i(u + ia))^2 - 2a\, i(u + ia) - \mu$ gives

$$C_j = -h_{j+2} - 2ah_{j+1} - \mu h_j, \qquad h_{j+1} = 0 \implies \mathcal{L}_j = \mu h_j^2 + h_j C_j. \tag{5.3}$$

Thus the contour drift disappears at exactly the original stationary constraint. No derivative of an exponentially normalized observation has replaced that constraint.

Proposition 5.2 (A sufficient residual cone)**.** *If $\eta > 0$ and*

$$\Delta := \eta^2 E - C_0^2 - C_1^2 \geq 0, \tag{5.4}$$

then the respective stationary sets satisfy

$$\mathcal{L}_0 \geq (\mu - \eta)E \quad (h_1 = 0), \qquad \mathcal{L}_1 \geq -\frac{\eta^2}{4\mu}E \quad (h_2 = 0). \tag{5.5}$$

A sufficient common negative-loss coefficient is $K = \eta^2/(4\mu)$. It is locally bounded wherever the prescribed parameters are locally bounded and μ stays positive.

Proof. At $h_1 = 0$, $E = h_0^2$ and $|C_0| \leq \eta\sqrt{E}$, so the first inequality follows from (5.3). At $h_2 = 0$, complete the square in $\mu h_1^2 - \eta|h_1|\sqrt{E}$ to obtain the second. Finally $\eta - \mu \leq \eta^2/(4\mu)$ because $(\eta - 2\mu)^2 \geq 0$. No division by E occurs. □

For use with exact algebra or interval certificates, the same argument has the following identities before stationarity is imposed:

$$\begin{aligned} 2\eta[\mathcal{L}_0 + (\eta - \mu)E] &= (C_0 + \eta h_0)^2 + C_1^2 + \Delta \\ &\quad + 2\eta h_1[(1 - \mu + \eta/2)h_1 + 2ah_0], \end{aligned} \tag{5.6}$$

$$4\mu\mathcal{L}_1 + \eta^2 E = (2\mu h_1 + C_1)^2 + C_0^2 + \Delta + 4\mu h_2(h_2 + 2ah_1). \tag{5.7}$$

Each final term is a multiple of the genuine stationary derivative. The cone is sufficient, but a one-sided bound for $h_j C_j$ can be substantially weaker and therefore more useful.

For example, suppose paid normalized approximations give $Z_j = e^{\pi x/8}h_j = p + \delta$ and $e^{\pi x/8}C_j = q + \epsilon'$, where $|\delta| \leq \varepsilon_0$, $|\epsilon'| \leq \zeta_0$. At $h_{j+1} = 0$, expansion of (5.3), retaining $\mu\delta^2 \geq 0$, gives

$$e^{\pi x/4}\mathcal{L}_j \geq \mu p^2 + pq - |2\mu p + q|\varepsilon_0 - |p|\zeta_0 - \varepsilon_0\zeta_0. \tag{5.8}$$

Every truncation, quadrature, and parameter error must be included in the two bounds. The retained product pq preserves its sign.

5.3 One carrier for all four derivatives

There is an exact arithmetic reduction behind the centered residual. Put $d_n = \frac{1}{2}\log n$. The source summands obey

$$\phi_n(z) = n^{-1/2}\phi_1(z + d_n). \tag{5.9}$$

For $v \geq d_n$, define

$$A_n(t, x; v) = n^{-1/2}\exp\{td_n^2 - 2td_n(v + ia) - ixd_n\}. \tag{5.10}$$

Changing variables $v = u + d_n$ in the complete shifted source yields

$$h_j = e^{-ax}\Re\int_0^\infty e^{t(v+ia)^2}\phi_1(v + ia)e^{ixv}\sum_{n \leq e^{2v}} A_n(t, x; v)[i(v + ia - d_n)]^j\, dv. \tag{5.11}$$

In particular $j = 0, 1, 2, 3$ uses only the first four logarithmic moments of the same arithmetic weights. The staircase cutoff $n \leq e^{2v}$ is part of the identity. It is not an independently chosen truncation.

To justify the rearrangement, let $0 \le t \le \tau$. Since $0 \le d_n \le v$, we have $|A_n| \le n^{-1/2}$ and $\sum_{n \le e^{2v}} n^{-1/2} \le 2e^v$. Therefore the absolute integrand in (5.11), before e^{-ax}, is bounded by

$$2(2\pi^2 + 3\pi)(v+a)^j e^{\tau v^2 + 10v - \pi \cos(4a) e^{4v}}. \tag{5.12}$$

This is integrable. It also gives a full integration tail, stated in Appendix A; no unretained index is omitted from that bound.

The common oscillatory exponential in the original n-th channel has phase $(x+2ta)u - \pi n^2 \sin(4a) e^{4u}$. Its critical location is

$$b_n = \frac{1}{4} \log \frac{x+2ta}{4\pi n^2 \sin(4a)} = b_1 - d_n. \tag{5.13}$$

Only nonnegative locations belong to the integration half-line. This observation supplies a centering choice; it does not assert first-channel dominance or a full stationary-phase approximation. Taking $b = b_1$, the multiplier defining C_j becomes

$$(v-d_n)^2 - b^2 = \underbrace{-b \log n + \tfrac{1}{4} \log^2 n}_{\delta_n} + 2(b-d_n)(v-b) + (v-b)^2. \tag{5.14}$$

Thus the remaining signed residual involves logarithmic weights of the same arithmetic sum, rather than unrelated errors for each derivative. At $v = b$ those weights have the transparent form

$$A_n = n^{-1/2 - tb - i(x/2+ta)} e^{t \log^2 n/4}. \tag{5.15}$$

The factors A_n are complex and their products retain arithmetic phases. A positive-weight covariance inequality does not apply to them.

For perspective, if $b > 0$ the frozen-parameter balanced quaternion $W = h_0 + \mathbf{j}(h_1 + ah_0)/b$ obeys

$$W_t = (b^2 - a^2 - 2ab\mathbf{j})W + R, \qquad R = C_0 + \mathbf{j}(C_1 - aC_0)/b. \tag{5.16}$$

The skew term rotates W without changing its norm. Equation (5.16) separates the expected oscillatory motion from the residual, whereas (5.3) keeps the precise scalar conditions needed for a sign. If a, b vary with time, the additional term is $\mathbf{j}(a_t \Re W/b - b_t \Im_{\mathbf{j}} W/b)$; all cell computations below freeze them before differentiating.

5.4 Staircase derivatives and exact entry terms

Put $T = x/2 + ta$ and, for integers $r \ge 0$,

$$B_r(v) = \sum_{n \le e^{2v}} (\log n)^r A_n(t,x;v), \qquad D_j(v) = \sum_{n \le e^{2v}} A_n(t,x;v)[i(v+ia-d_n)]^j.$$

For $r = 0$ the logarithmic weight is one, including at $n = 1$. On $v > 0$, distributional differentiation gives

$$B_r' = -tB_{r+1} + \sum_{n \ge 2} (\log n)^r A_n(t,x;d_n)\delta(v-d_n). \tag{5.17}$$

Between entry points,

$$D_j' = -2t(v+ia)D_j - 2itD_{j+1} + ijD_{j-1}, \tag{5.18}$$

where the last term is omitted for $j = 0$. The jump at d_n is $A_n(t,x;d_n)(-a)^j$. If $K(v) = e^{t(v+ia)^2}\phi_1(v+ia)e^{ixv}$, its full contribution is

$$K(d_n)A_n(t,x;d_n)(-a)^j = e^{-ta^2}\phi_n(ia)(-a)^j. \tag{5.19}$$

Indeed $\partial_v A_n = -2td_nA_n$ and $d_n = v + ia + i[i(v + ia - d_n)]$ give the smooth recurrence; direct substitution and (5.9) give the jump formula. At $v = 0$ the index $n = 1$ is already present and belongs to the lower endpoint of any integration by parts. Every entering-index phase in (5.19) cancels in x. These terms cannot be assigned an artificial oscillatory gain.

The oscillatory block estimate in Section 2 can also be applied to this exact carrier. On a block $M < n \le R(v)$, where $R(v) = \min(2M, e^{2v})$, the logarithmic-moment weight is

$$w_r(y) = (\log y)^r e^{t\log^2 y/4} y^{-1/2-tv}.$$

For $M \ge e^{2r}$ it is decreasing, since $y(\log w_r)' \le r/\log y - 1/2 \le 0$; for $r = 0$ this holds for all $M \ge 1$. Abel summation therefore gives, when $|T| \ge 1$,

$$\left| \sum_{M<n\le R(v)} w_r(n) n^{-iT} \right| \ll w_r(M)(|T|^{1/6} M^{1/2} + |T|^{-1/6} M).$$

For other blocks use the full total variation of w_r, and for $|T| < 1$ use the trivial bound. The actual shortened endpoint remains $R(v)$; these complex block estimates do not imply a signed product bound.

5.5 The complete paired observations

Define the staircase measure space and its coordinates by

$$\begin{aligned} &\Omega = \{(v,n) : n \ge 1,\ v \ge d_n\}, \qquad d\nu = dv \times \text{counting measure}, \\ &y_\alpha = v - d_n, \qquad q_\alpha = -a + iy_\alpha, \\ &W_\alpha = e^{t(v+ia)^2} \phi_1(v + ia) e^{ixv} A_n(t, x; v), \qquad \alpha = (v, n). \end{aligned} \tag{5.20}$$

Then $h_k = e^{-ax}\Re \int_\Omega W_\alpha q_\alpha^k \, d\nu(\alpha)$. The two factors in the following integrals have independent coordinates and independent indices.

Proposition 5.3 (Paired Laguerre and residual forms)**.** *For $j = 0, 1$,*

$$\begin{aligned} \mathcal{L}_j = \frac{e^{-2ax}}{4} \Re \iint_{\Omega^2} [&(y_\alpha + y_\beta)^2 W_\alpha \overline{W_\beta} (q_\alpha \overline{q_\beta})^j \\ &+ (y_\alpha - y_\beta)^2 W_\alpha W_\beta (q_\alpha q_\beta)^j] \, d\nu(\alpha) d\nu(\beta). \end{aligned} \tag{5.21}$$

Writing

$$J_j(\alpha, \beta) = W_\alpha \overline{W_\beta} (q_\alpha \overline{q_\beta})^j + W_\alpha W_\beta (q_\alpha q_\beta)^j, \tag{5.22}$$

the centered product is

$$h_j C_j = \frac{e^{-2ax}}{4} \Re \iint_{\Omega^2} (y_\alpha^2 + y_\beta^2 - 2b^2) J_j(\alpha, \beta) \, d\nu(\alpha) d\nu(\beta). \tag{5.23}$$

Before imposing stationarity, the exact difference is

$$\mathcal{L}_j - \mu h_j^2 - h_j C_j = h_{j+1}(h_{j+1} + 2a h_j). \tag{5.24}$$

Proof. Use $\Re A \Re B = \frac{1}{2}\Re(A\overline{B} + AB)$ and symmetrize in α, β. The Hermitian difference in the Laguerre form is

$$q_\alpha \overline{q_\beta} - \tfrac{1}{2}(q_\alpha^2 + \overline{q_\beta}^2) = -\tfrac{1}{2}(q_\alpha - \overline{q_\beta})^2 = \tfrac{1}{2}(y_\alpha + y_\beta)^2;$$

the bilinear difference is $-\frac{1}{2}(q_\alpha - q_\beta)^2 = \frac{1}{2}(y_\alpha - y_\beta)^2$. Conjugate-swapping the Hermitian term preserves its real integral. The same polarization applied to the real multiplier $y^2 - b^2$ gives (5.23), and (5.3) gives (5.24). The majorant (5.12) has every polynomial moment integrable. Together with $(y \pm z)^2 \le 2(y^2 + z^2)$, this proves absolute convergence of each displayed integral and justifies Fubini. □

The squared frequencies in (5.21) do not give the phase of either amplitude product a sign. As a finite algebra control, two real signed atoms at frequencies $1, 2$, with weights $1, -1/2$ and $a = 0$, give $(h_0, h_1, h_2) = (1/2, 0, 1)$ and $\mathcal{L}_0 = -1/2$. The diagonal contributions sum to 2 and the ordered cross terms to $-5/2$. This control lies outside the theta class and positive-contour hypotheses; it explains why retaining only diagonal pairs loses the observation.

5.6 A complete omitted-pair payment

For a measurable retained set $A \subset \Omega$, denote its jets by h_k^A and set $\mathcal{L}_j^A = (h_{j+1}^A)^2 - h_j^A h_{j+2}^A$. Define absolute masses before the exterior exponential normalization:

$$M_k^A = \int_A |W_\alpha||q_\alpha|^k \, d\nu, \qquad T_k^A = \int_{\Omega\setminus A} |W_\alpha||q_\alpha|^k \, d\nu.$$

Then

$$|\mathcal{L}_j - \mathcal{L}_j^A| \le e^{-2ax}(M_j^A T_{j+2}^A + M_{j+2}^A T_j^A + T_j^A T_{j+2}^A). \tag{5.25}$$

Indeed, including the factor $1/4$ in (5.21), the absolute sum of its two kernels is at most

$$\tfrac{1}{2}(|W_\alpha||q_\alpha|^{j+2}|W_\beta||q_\beta|^j + |W_\alpha||q_\alpha|^j|W_\beta||q_\beta|^{j+2}),$$

using $(y+z)^2 + (y-z)^2 = 2(y^2+z^2)$ and $y^2 \le |q|^2$. Integrating over $A \times A^c$, $A^c \times A$, and $A^c \times A^c$ gives the claim, including all omitted mixed and omitted–omitted pairs. No bound for T_{j+1}^A is required.

For $A = \{(v, n) \in \Omega : v \le V,\ n \le N\}$, use the complete index payment and the shared-coordinate integration payment in Appendix A, removing their e^{-ax} normalization before using them for T_k^A. Overlap may be overpaid. Retained quadrature and parameter variation require their own errors. The absolute payment (5.25) is not a relative bound through an unknown zero of E.

6 The bounded region: a complete certificate through height 104

The bounded calculation supplies a genuine norm gap, so it excludes collisions directly. Its role is to join the central analytic estimates to the large-height theorem, while leaving the intervening and small-$t \log|x|$ regions explicitly visible.

Theorem 6.1 (Certified closed bounded region)**.** *For the complete theta source,*

$$|F(t,x)| = \sqrt{h_0(t,x)^2 + h_1(t,x)^2} > 10^{-16} \quad (0 \le t \le 1/2,\ |x| \le 104). \tag{6.1}$$

On this rectangle, $h_1 = 0$ implies $\mathcal{L}_0 > 0$, and $h_2 = 0$ implies $\mathcal{L}_1 > 0$. The norm bound combines an analytic central estimate with the validated interval records described below.

6.1 A central estimate with no theta truncation

Proposition 6.2 (Central norm gap)**.** *For $0 \le t \le 1$ and $|x| \le 8$, $|F(t,x)| > 1/120000$. Both stationary signs in Theorem 6.1 hold on this larger time interval.*

Proof. Put $w_t(u) = e^{tu^2}\Phi(u)$ and $M_k(t) = \int_0^\infty u^k w_t(u)\,du$. The complete source obeys $0 < \Phi(u) < 4\pi^2 e^{9u-\pi e^{4u}}$ for $u \ge 0$: after dropping each summand's negative term, the ratio of successive $n^4 e^{-\pi e^{4u} n^2}$ terms is at most $16e^{-3\pi} < 1/2$. The following elementary lower cores and upper tails hold uniformly for $0 \le t \le 1$:

core $[0,d]$	$\Phi(u) > c_d$	tail $u \ge b$	$w_t(u) < A_b e^{-q_b(u-b)}$
$d = 1/10$	$6/25$	$b = 7/25$	$(1/20)e^{-28(u-b)}$
$d = 3/20$	$3/25$	$b = 3/10$	$(1/49)e^{-32(u-b)}$
$d = 1/5$	$1/20$		

(6.2)

For completeness, the lower bounds follow from $\phi_1(u) = e^u g(\pi e^{4u})$ with $g(q) = q(2q-3)e^{-q}$ and $g'(q) = -(2q-1)(q-3)e^{-q}$. On the three cores the respective arguments are less than $33/7, 6, 36/5$. The bounds $e < 11/4$, $e^{2/7} > 9/7$, and $e^{1/5} < 5/4$ give the three stated rational lower bounds for g. For the tails, $\psi_t(u) = 9u + tu^2 - \pi e^{4u}$ is strictly concave. At $b = 7/25$ one has $\psi_t(b) < -69/10$, $\psi_t'(b) < -28$; at $b = 3/10$ one has $\psi_t(b) < -38/5$, $\psi_t'(b) < -32$. Together with $4\pi^2 < 40$, $e^{69/10} > 800$, and $e^{38/5} > 1960$, these prove the full upper tails in (6.2). All comparisons use rational Taylor bounds for the indicated exponentials; in particular they do not truncate the theta source.

Integrating $u^2(1 - 25u^2/2)$ over the first positive core and paying its entire negative tail with $b = 7/25$ gives

$$M_2 - \frac{25}{2}M_4 > \frac{37}{500000} - \frac{5}{8}\frac{3497259}{33614000000} = \frac{2413193}{268912000000} > \frac{1}{120000}. \tag{6.3}$$

Indeed the polynomial is nonnegative up to b and is bounded below on the tail by $-(25/2)u^2(u^2-b^2)$. Likewise, with $f(u) = u^4(1-32u^2/3)$ and $b = 3/10$, the three successive cores and the entire negative tail give

$$M_4 - \frac{32}{3}M_6 > \frac{111799}{33600000000} - \frac{32}{147}\frac{85837761}{6710886400000} = \frac{2092283587}{3853516800000000} > \frac{1}{2000000}. \tag{6.4}$$

The positive term is $(6/25)\int_0^{1/10} f + (3/25)\int_{1/10}^{3/20} f + (1/20)\int_{3/20}^{1/5} f$; the tail integral is $\int_b^\infty u^4(u^2 - b^2)e^{-32(u-b)}\,du$. Thus both displayed differences are exact rational integrations.

The global inequalities $\sin y \ge y - y^3/6$ for $y \ge 0$ and $\cos y \ge 1 - y^2/2$ now imply, for $0 < x \le 8$,

$$-h_1 \ge x(M_2 - x^2 M_4/6) > x/120000, \qquad h_3 \ge x(M_4 - x^2 M_6/6) > x/2000000.$$

For $|x| \le 1$, integrate the first core using $\cos(xu) \ge 199/200$, discard the nonnegative middle interval $[1/10, 7/25]$, and pay the entire remaining tail. This gives

$$H(t,x) > \frac{597}{25000} - \frac{1}{560} = \frac{7733}{350000} > \frac{1}{50}.$$

The value bound covers $|x| \le 1$, and the slope bound covers $1 \le |x| \le 8$, proving the norm gap by evenness. The first stationary set is exactly $x = 0$, where $\mathcal{L}_0 = H(t,0)M_2 > 0$. For $0 < |x| \le 8$, $\mathcal{L}_1 = h_2^2 - h_1 h_3 > x^2/240000000000$; at $x = 0$ one has $h_2 = -M_2 < 0$, so the second stationary set is absent. □

6.2 What the interval certificates enclose

The records use Arb/Acb interval arithmetic to enclose the complete physical jets on closed space–time cells. Each enclosure includes the returned finite-quadrature radius, every omitted theta index, the complete integration tail, rounding, and spatial and temporal Taylor remainders. The older compensated calculation also pays the complete Gaussian transforms and their truncation remainders. Finite compensated channels are transported using the raw heat equation followed by their explicit time-dependent compensation; they are not assigned a homogeneous heat equation.

For a heat-transported sequence Y_r, a cell with half-widths d_t, d_x uses

$$Y_r(t_c+\tau, x_c+\eta) = \sum_{\ell=0}^{q-1}\sum_{m=0}^{p-1} \frac{(-\tau)^\ell \eta^m}{\ell! m!} Y_{r+2\ell+m}(t_c, x_c) + \mathcal{R}_r.$$

If M_k bounds the complete kth derivative on the cell, Taylor's theorem first in time and then in space gives

$$|\mathcal{R}_r| \le \sum_{\ell=0}^{q-1} \frac{d_t^\ell}{\ell!}\frac{d_x^p}{p!} M_{r+2\ell+p} + \frac{d_t^q}{q!} M_{r+2q}. \tag{6.5}$$

The higher bands sum the retained theta coefficients before applying the common Taylor monomials. This preserves cancellation between different indices while adding every positive remainder payment.

Stationary decisions concern genuine derivatives. If $h_k = p_k + r_k$ with $|r_k| \le \varepsilon_k$, then at $h_{j+1} = 0$ one has

$$\mathcal{L}_j \ge p_{j+1}^2 - p_j p_{j+2} - (\varepsilon_{j+1}^2 + |p_j|\varepsilon_{j+2} + |p_{j+2}|\varepsilon_j + \varepsilon_j\varepsilon_{j+2}). \tag{6.6}$$

Here $p_{j+1}^2 \le \varepsilon_{j+1}^2$ follows from genuine stationarity; the square payment has a negative sign. The products retain all cross terms between retained channels. A cell is discharged either by exclusion of the stationary derivative or by a strictly positive paid signed lower bound. Monotone endpoint exclusions, where used, employ the next physical derivative and uniform endpoint signs. No zero of an ordinary derivative of an exponentially rescaled function is substituted.

Independently, if I_0, I_1 enclose the genuine h_0, h_1 throughout a cell, then

$$E \ge \operatorname{dist}(0, I_0)^2 + \operatorname{dist}(0, I_1)^2. \tag{6.7}$$

Exact rational comparisons and downward square-root rounding turn these energy floors into the reported norm floors. No division by an unproved norm is used.

6.3 Coverage, the centered-residual pilot, and reproducibility

Every numerical band covers the full closed time interval $[0, 1/2]$ using eight slabs $[r/16, (r+1)/16]$. The records give the following conservative norm bounds; the central row is analytic.

Positive-height band	Cells	Precision	Proved lower bound for $\lvert F\rvert$
$[0, 8]$	—	exact estimates	$1/120000$
$[7, 41]$	272	256 bits	$9/5000000$
$[40, 100]$	510	256 bits	10^{-16}
$[100, 104]$	32	192 bits	$4 \cdot 10^{-16}$

The 510 cells comprise 160, 170, and 180 cells in $[40, 60]$, $[60, 80]$, and $[80, 100]$. Their exact norm floor is

$$\frac{22271745300542300149807}{85070591730234615865843651857942052864} > 10^{-16}.$$

Exact rational cover checks include the spatial subdivisions, all shared boundaries, and both time endpoints. The spatial overlaps $[7,8]$, $[40,41]$, and the shared boundary $x = 100$, together with evenness, prove (6.1).

The last band also certifies the centered-residual mechanism. On each of its 32 cells, freeze

$$a = \frac{\pi}{16}, \qquad b = \frac{1}{4}\log\frac{x_c + 2t_c a}{4\pi\sin(4a)}, \qquad \mu = a^2 + b^2, \qquad C_j = -h_{j+2} - 2ah_{j+1} - \mu h_j.$$

The complete even source is shifted before theta indices are separated. With eight retained indices, integration cutoff $3/2$, spatial order 12 and time order 8, the computation encloses all center jets through order 28 and complete derivative majorants through order 29. Centered coefficients are combined before Taylor transport, with the parameters frozen on the cell. The cell-constant normalization $e^{\pi x_c/8}$ rescales each physical jet equally. For normalized enclosures P_0, P_1, Q_0, Q_1 of h_0, h_1, C_0, C_1, the actual decisions are

$$E_{\mathrm{low}} = \operatorname{dist}(0,P_0)^2 + \operatorname{dist}(0,P_1)^2 > 0, \qquad E_{\mathrm{low}}/9 - \sup|Q_0|^2 - \sup|Q_1|^2 > 0.$$

Thus $C_0^2 + C_1^2 \le E/9$ and $\mu \ge 2/5$ throughout every cell. The exact stationary identity $\mathcal{L}_j = \mu h_j^2 + h_j C_j$ at $h_{j+1} = 0$ yields

$$\mathcal{L}_0 \ge E/15 \quad (h_1 = 0), \qquad \mathcal{L}_1 \ge -5E/72 \quad (h_2 = 0).$$

The second strict stationary sign in Theorem 6.1 is a separate validated signed decision; it is stronger than the bounded-loss conclusion supplied by this cone. At each level the band has 24 derivative-exclusion cells and eight strict signed cells, with no claim that a signed cell alone proves existence or uniqueness of a stationary branch.

Appendix C identifies the retained drivers, records, exact auditors, and earlier independent internal replays. In particular, the higher-height auditor checks 6682 assertions and the centered-band auditor checks 547. Those checks concern arithmetic, decisions, complete coverage, and source bindings. The complete-source contour bounds used for the latest band are proved in Appendix A. The computations establish the stated closed bounded region; they do not extend it to arbitrary heights.

7 One-sided stationary estimates and their obstructions

The outer theorem and the finite cover give complementary information. For each fixed $\varepsilon > 0$, the unresolved positive-time domain is contained in the transition strip $104 < |x| < X_\varepsilon$ and the high-height inner region

$$0 < t \le \tfrac{1}{2}, \qquad |x| \ge X_\varepsilon, \qquad t\log(|x|/(4\pi)) < 3 + \varepsilon. \tag{7.1}$$

The threshold-three assertion is asymptotic: it does not supply a usable numerical value of X_ε. The explicit threshold-five bound proved separately is not a numerical implementation of threshold three. The remaining analytic problem is a signed estimate in (7.1), with constants regular through a candidate positive-time collision. The following obstructions delimit that problem.

7.1 Absolute dominance and pointwise positivity are too strong

At the exact common-carrier coordinate v, put $M = \lfloor e^{2v} \rfloor$. For $\lceil M/2 \rceil \le n \le M$,

$$td_n^2 - 2td_n v = t[(d_n - v)^2 - v^2] \ge -tv^2.$$

Consequently

$$\sum_{n\le e^{2v}} |A_n| \ge (M/2-1)M^{-1/2}e^{-tv^2}. \tag{7.2}$$

With fixed a, $v=b_1$, and $t=\vartheta/\log x$ for fixed $0\le\vartheta<4$, this grows as a positive constant times $x^{1/4-\vartheta/16}$, whereas $|A_1|=1$. Absolute first-channel dominance therefore fails in this scaling. Cancellation within the sum and after integration remains available; indeed it is essential to the improved outer theorem.

Pointwise positivity is also unavailable. For fixed $v=b$, write

$$D_j=\sum_{n\le e^{2b}} A_n(t,x;b)[-a+i(b-d_n)]^j, \qquad p_j=\Re(UD_j),$$

where U is the unit phase of $e^{t(b+ia)^2}\phi_1(b+ia)e^{ixb}$. Put

$$\mathfrak{H}_j=|D_{j+1}|^2-\Re(D_j\overline{D_{j+2}}), \qquad \mathfrak{W}_j=D_{j+1}^2-D_jD_{j+2}.$$

Then

$$2(p_{j+1}^2-p_jp_{j+2})=\mathfrak{H}_j+\Re(U^2\mathfrak{W}_j). \tag{7.3}$$

The stronger phase-independent condition $\mathfrak{H}_j\ge|\mathfrak{W}_j|$ fails for the actual common-carrier sums. With $a=\pi/16$ and $b=b_1$ from (5.13), retained Arb/Acb enclosures give

x	t	$\mathfrak{H}_0-\lvert\mathfrak{W}_0\rvert$ (approximate value)
211	$1/\log 211$	-0.0277198
4898.7	$2/\log 4898.7$	-1.26105

At the second point the actual phase also gives $p_1^2-p_0p_2\approx-0.01099197<0$. The stored enclosures are strictly negative; Appendix C identifies their replay. These are pointwise integrand forms, not the integrated Laguerre forms, and no physical stationarity is asserted at these points. The paired identities in Section 5 retain correlations between distinct coordinates as well as distinct indices. A proof requiring every integrand to be positive would discard that possibility. The staircase boundary identity (5.19) must likewise be retained: entering terms have an exact cancellation of their arithmetic x-phase, not a freely discardable endpoint error.

7.2 A sufficient split that preserves favorable curvature

Throughout this section let $0<a<\pi/8$, prescribe a real b, and put $\mu=a^2+b^2$. At each point the centered observations are

$$C_k=-h_{k+2}-2ah_{k+1}-\mu h_k, \qquad k\ge0. \tag{7.4}$$

Whenever these observations are differentiated or compared at different times below, a,b are held fixed. The exact identity is

$$\mathcal{L}_j=\mu h_j^2+h_jC_j+h_{j+1}(h_{j+1}+2ah_j).$$

Thus the remaining target can be stated as

$$\boxed{h_jC_j\ge-\mu h_j^2-K_jE \quad \text{on } h_{j+1}=0, \qquad j=0,1,} \tag{7.5}$$

where $K_j\ge0$ is prescribed without dividing by an unknown energy and is locally bounded through every candidate positive-time collision. This is precisely the stationary loss condition in Section 3.

Lemma 7.1 (Favorable curvature and a residual square)**.** *Prescribe an exact split $C_j = G_j + R_j$ and parameters $0 < \kappa \le \mu$, $\rho \ge 0$. Define*

$$\Delta_j = 4\kappa h_j G_j + \rho^2 E - R_j^2. \tag{7.6}$$

Without imposing stationarity,

$$\begin{aligned} 4\kappa\mathcal{L}_j + \rho^2 E = (R_j + 2\kappa h_j)^2 + \Delta_j + 4\kappa(\mu-\kappa)h_j^2 \\ + 4\kappa h_{j+1}(h_{j+1} + 2a h_j). \end{aligned} \tag{7.7}$$

In particular, $\Delta_j \ge 0$ on $h_{j+1} = 0$ implies $\mathcal{L}_j \ge -\rho^2 E/(4\kappa)$. This loss is locally bounded if ρ is locally bounded and κ is locally bounded below by a positive constant.

Proof. Expand the square and use $G_j + R_j = C_j$. All terms are polynomial in the physical jets, so neither $E > 0$ nor division by a jet is required. On stationarity the last term vanishes and the other displayed squares and $4\kappa(\mu-\kappa)h_j^2$ are nonnegative. □

The split must expose an independently estimable source contribution. For the exact domain and fields of (5.20), choose a prescribed real multiplier $m_j(\alpha)$ with at most polynomial growth in y_α, and set

$$\begin{aligned} G_j &= e^{-ax}\Re \int_\Omega m_j(\alpha) W_\alpha q_\alpha^j \, d\nu(\alpha), \\ R_j &= e^{-ax}\Re \int_\Omega r_j(\alpha) W_\alpha q_\alpha^j \, d\nu(\alpha), \qquad r_j(\alpha) = y_\alpha^2 - b^2 - m_j(\alpha). \end{aligned} \tag{7.8}$$

A proper source filter takes $m_j = (y_\alpha^2 - b^2)\mathbf{1}_A$, with the retained set A intersected with the original staircase. The filter and its parameters are fixed before evaluating the physical observations. For real m_j, r_j, the polarization (5.22) gives the exact scalar obligation

$$\begin{aligned} \Delta_j = \frac{e^{-2ax}}{2}\Re \iint_{\Omega^2} \big[\{2\kappa(m_j(\alpha) + m_j(\beta)) - r_j(\alpha) r_j(\beta)\} J_j(\alpha,\beta) \\ + \rho^2 (J_0(\alpha,\beta) + J_1(\alpha,\beta))\big] \, d\nu(\alpha)\, d\nu(\beta). \end{aligned} \tag{7.9}$$

Indeed $h_j G_j$ is $e^{-2ax}/4$ times the real part of the integral of $(m_j(\alpha) + m_j(\beta))J_j$, whereas R_j^2 has coefficient $e^{-2ax}/2$ and integrand $r_j(\alpha) r_j(\beta) J_j$. The complete source majorants justify these identities and their tail payments; they do not prove the sign in (7.9). The formula requires real multipliers. A complex filter requires its own conjugated polarization or an independent physical-field identity.

Even a quotient-free split can be vacuous: choosing $m_j = y_\alpha^2 - b^2 + 2\kappa$ makes $R_j = -2\kappa h_j$ and cancels the first square in (7.7). With $\kappa = \mu$, the resulting slack condition only restates the target and supplies no theta estimate.

7.3 Collision controls and the limitation of the square criterion

For the backward-heat polynomial $P(t,y) = y^2 - d^2 - 2t$, put $D = d^2 + 2t > 0$. Its Laguerre forms are $2y^2 + 2D > 0$ and 4. At $y = 0$, $h_0 = -D$, $h_1 = 0$, $E = D^2$, $C_0 = -2 + \mu D$, and $C_1 = -4a$. The whole-residual cone would require

$$\eta \ge \frac{\sqrt{(2-\mu D)^2 + 16a^2}}{D}. \tag{7.10}$$

For bounded a, μ, this grows without bound as the pair closes, although the required negative loss is zero. The prescribed split $G_0 = -2$, $R_0 = \mu D$ instead gives

$$\Delta_0 = 8\kappa D + (\rho^2 - \mu^2)D^2.$$

It accepts $\rho = 0$ whenever $0 < D \le 8\kappa/\mu^2$, preserving the large favorable curvature. This control is not a theta counterexample; growth as $t \downarrow 0$ is also compatible with local boundedness at each positive time.

The same algebra rejects adverse collision predecessors. For $P = y^2 - 2(t - T)$ at $y = 0$, $t = T - s < T$, the split $G_0 = -2$, $R_0 = -2\mu s$ yields

$$\Delta_0 = -16\kappa s + 4(\rho^2 - \mu^2)s^2, \qquad \mathcal{L}_0 = -4s, \qquad E = 4s^2.$$

For $P = y^3 - 6(t - T)y$ at $y = 0$, $t = T - s < T$, the second stationary set $h_2 = 0$ has $h_1 = 6s$, $h_3 = 6$. Taking $G_1 = -6$, $R_1 = -6\mu s$ gives

$$\Delta_1 = -144\kappa s + 36(\rho^2 - \mu^2)s^2, \qquad \mathcal{L}_1 = -36s, \qquad E = 36s^2.$$

Both slacks are negative for sufficiently small $s > 0$ when ρ is bounded and κ stays positively bounded below. The second stationary set cannot be omitted.

There is also a precise restriction at the candidate collision itself: if $h_0 = h_1 = 0$, then $\Delta_0 = -R_0^2$. The square criterion therefore forces $R_0 = 0$ exactly; if also $h_2 = 0$, it forces $R_1 = 0$. A small absolute bound on an omitted source does not imply this vanishing. Thus Lemma 7.1 is a sufficient criterion with an extra cancellation obligation, not a solution of the remaining theta problem.

Nor can its sign be justified by arbitrary-source positivity. On $h_1 = 0$ the arbitrary-jet form $\mathcal{L}_0 + KE = -h_0 h_2 + K h_0^2$ is indefinite for every finite K; on $h_2 = 0$ the form is $-h_1 h_3 + K(h_0^2 + h_1^2)$ and is likewise indefinite. Terms in the stationary ideal vanish there. Any useful positivity argument must therefore use relations of the actual theta source.

7.4 Direct product absorption

The residual-square condition can be replaced by a one-sided product condition which imposes no residual vanishing at $E = 0$.

Lemma 7.2 (Fixed-fraction absorption)**.** *For a prescribed split $C_j = G_j + R_j$, let $0 \le \lambda \le 1$ and $K_j \ge 0$. On the genuine stationary set $h_{j+1} = 0$, suppose*

$$h_j G_j \ge 0, \qquad h_j(R_j + \lambda G_j) \ge -K_j E. \tag{7.11}$$

Then $\mathcal{L}_j \ge -K_j E$.

Proof. On this set,

$$\mathcal{L}_j = \mu h_j^2 + (1 - \lambda)h_j G_j + h_j(R_j + \lambda G_j) \ge -K_j E.$$

□

At a first-pair zero the displayed products vanish automatically, but the estimate must still hold with a locally bounded prescribed loss in a neighborhood of that point. A favorable retained product alone does not suffice. The finite results below use $\lambda = 1/2$ and $K_0 = K_1 = 0$, without fitting a loss to an observed energy floor.

8 Proper source filters and complete theta correlations

We first test the preceding mechanisms on populated stationary branches inside the existing height-104 cover. We then prove why fixed sharp filters cannot support uniformly small separate payments at arbitrary heights, and derive a complete-source alternative. The finite tests do not extend the bounded cover or settle (7.1).

8.1 Two certified stationary branches and a signed comparison

Fix

$$a = \pi/16, \qquad b = 1/4, \qquad \mu = a^2 + b^2, \qquad \lambda = 1/2, \qquad K_0 = K_1 = 0. \tag{8.1}$$

Use the constant normalization $Z_k = e^{\pi x_c/8} h_k$, with $x_c = 101.5$ on $[101, 102]$ and $x_c = 103.5$ on $[103, 104]$. Hats denote the same normalization of centered observations. The complete-source certificate and paid endpoint Taylor transport give, for every $t \in [0, 1/2]$,

$$\begin{aligned} &Z_1(t, 101) > 37, \qquad Z_1(t, 102) < -20, \qquad Z_2 < 0 \quad (101 \le x \le 102), \\ &Z_2(t, 103) < -26, \qquad Z_2(t, 104) > 14, \qquad Z_3 > 0 \quad (103 \le x \le 104). \end{aligned} \tag{8.2}$$

The intermediate value theorem and strict monotonicity therefore prove exactly one $h_1 = 0$ point in $(101, 102)$ and exactly one $h_2 = 0$ point in $(103, 104)$ at each time. The implicit function theorem gives smooth branches. These are stationary points of the physical derivatives, not derivatives of a variable normalization.

A first signed comparison uses the full residual on the first branch. Exact rational extraction from the eight stored closed time slabs gives

$$\begin{gathered} Z_0 > 0, \qquad \widehat{C}_0 > 0, \qquad Z_0 \widehat{C}_0 > 6000, \qquad \widehat{C}_0 - \mu Z_0 > 7/5, \\ \mu Z_0^2 + Z_0 \widehat{C}_0 > 9400, \qquad \mu^2 Z_0^2 - \widehat{C}_0^2 - \widehat{C}_1^2 < -78. \end{gathered} \tag{8.3}$$

Thus the signed condition holds with $K_0 = 0$, whereas every prescribed whole-residual cone budget $\eta \le \mu$ fails on this branch. The previously certified saddle-centered $\eta = 1/3$ cone still passes; the comparison concerns the different fixed centering (8.1), not a failure of every norm cone. Taking $G = C$, $R = 0$ here is a full-residual comparison, not a proper source split. The next experiment supplies such a split.

8.2 A proper filter absorbs an adverse complement on 64 cells

Before evaluating partial integrals, prescribe the four filters

$$(N, U) \in \{(1, 1), (1, 3/2), (2, 1), (2, 3/2)\}.$$

For $z = u + ia$, define

$$\begin{aligned} g_k(t, x) &= e^{-ax} \Re \sum_{n=1}^{N} \int_0^U (iz)^k e^{tz^2 + ixu} \phi_n(z)\, du, \\ G_j &= -g_{j+2} - 2a g_{j+1} - \mu g_j, \qquad R_j = C_j - G_j. \end{aligned} \tag{8.4}$$

This is precisely the real source multiplier $(u^2 - b^2)\mathbf{1}_{n \le N,\, 0 \le u \le U}$. Its shared-carrier domain is $n \le N$, $d_n \le v \le d_n + U$, so all entry boundaries are retained. Each of the eight time slabs $[r/16, (r+1)/16]$, $0 \le r \le 7$, is tested on both branches: four filters, two branches, and eight slabs give 64 closed-cell

tests. Even-index slabs were designated development and odd-index slabs holdout before evaluation; no filter, parameter, or loss was changed between them. Their interiors are disjoint and their common boundary times are covered.

Both values of U give the following conservative bounds on every corresponding slab. Products are normalized by $e^{\pi x_c/4}$.

N	j	$Z_j\widehat{G}_j$	$Z_j\widehat{R}_j$	$Z_j(\widehat{R}_j+\widehat{G}_j/2)$
1	0	>0	<0	$<-1{,}200{,}000$
1	1	>0	<0	$<-250{,}000$
2	0	>7800	$(-212,-120)$	>3800
2	1	>3800	$(-46,-25)$	>1900

Here $j=0$ means the $h_1=0$ branch and $j=1$ the $h_2=0$ branch. The two-index filters therefore pay a genuinely adverse complementary product with the prescribed half of a favorable retained product. Lemma 7.2 applies with the fixed zero loss. For one index the same prescribed absorption test actually fails on nonempty stationary sets: the negative upper bounds are not an interval precision failure. The large one-index terms nearly cancel in the complete residual.

For completeness, the certification encloses retained center derivatives through order 28 with 192-bit Arb quadrature, then uses space order 12 and time order 8 for transport. Fixed endpoints give $\partial_x g_k = g_{k+1}$ and $\partial_t g_k = -g_{k+2}$ exactly. The complete-source moment payments remain valid for the three masks $\mathbf{1}_A$, $1-\mathbf{1}_A$, and $1-\mathbf{1}_A/2$, all of which lie in $[0,1]$. Centered payments are bounded by $P_{j+2}+2aP_{j+1}+\mu P_j$ for each spatial or temporal remainder. The full C arrays already include quadrature error, all omitted theta indices, and the infinite integration tail. The complement is the exact difference $C-G$, and $C-G/2$ is formed before transport to preserve cancellation. On current stationarity its enclosure is intersected with $-h_{j+2}-\mu h_j - G_j/2$; the complement is treated similarly. The product always uses the full physical jet and an integrated observation, retaining all cross terms. No individual theta index is assigned the vanishing vertical contour integral of the complete source. The complete 64-case receipt, including failures, is identified in Appendix C.

8.3 The endpoint terms of a fixed sharp filter

Now fix $t\in(0,1/2]$, $0<a<\pi/8$, $b\ge 0$, $N\ge 1$, and $U>0$, independently of x. Write

$$\Phi_N(z)=\sum_{n=1}^{N}\phi_n(z),\qquad q=-a+iu=iz,\qquad f_{N,j}(u)=(u^2-b^2)q^j e^{tz^2}\Phi_N(z).$$

Then the retained residual in (8.4) is

$$G_j(x)=e^{-ax}\Re\int_0^U f_{N,j}(u)e^{ixu}\,du,\qquad R_j=C_j-G_j. \tag{8.5}$$

These fixed filtered fields still solve backward heat. The following effect comes from source integration by parts, not a heat-equation defect.

Lemma 8.1 (Complete endpoint expansion)**.** *For $x>0$,*

$$\begin{aligned} e^{ax}G_j(x)&=\frac{B_j(x)}{x}+\mathcal{E}_j(x),\qquad B_j(x)=\Im\{f_{N,j}(U)e^{ixU}-f_{N,j}(0)\},\\ |\mathcal{E}_j(x)|&\le\frac{M_{N,j}}{x^2},\qquad M_{N,j}=|f'_{N,j}(U)|+|f'_{N,j}(0)|+\int_0^U|f''_{N,j}(u)|\,du. \end{aligned} \tag{8.6}$$

In particular the lower coefficient is

$$-\Im f_{N,j}(0) = b^2(-a)^j e^{-ta^2}\Im\Phi_N(ia). \tag{8.7}$$

Proof. Twice integrating $\int_0^U fe^{ixu}\,du$ by parts gives

$$\frac{[fe^{ixu}]_0^U}{ix} - \frac{[f'e^{ixu}]_0^U}{(ix)^2} + \frac{1}{(ix)^2}\int_0^U f''e^{ixu}\,du.$$

Taking real parts uses $\Re(-iZ) = \Im Z$ and gives the stated bound. Substitution at $u = 0$ proves (8.7). □

For the complete theta source, evenness and Schwarz symmetry make $\Phi(ia)$ real. More generally its centered integrand obeys $f_j(-u) = \overline{f_j(u)}$. Its even derivatives at zero are real and its odd derivatives purely imaginary, so every lower-endpoint coefficient in the real integration-by-parts expansion vanishes. A finite raw-index sum need not have this symmetry. If instead a common cutoff $v \le V$ is used, retain $U_n = V - d_n > 0$ for every included index; the leading term becomes

$$\Im\left\{\sum_{\substack{n\le N\\ d_n<V}} f_{n,j}(U_n)e^{ixU_n} - \sum_{\substack{n\le N\\ d_n<V}} f_{n,j}(0)\right\}.$$

Here $f_{n,j}$ is defined with the individual term ϕ_n in place of Φ_N. The remainder is the sum of the bounds in (8.6). An index with $d_n = V$ has an empty interval and contributes zero, not a single unmatched endpoint. This is the same entry cancellation as (5.19).

The upper endpoint is provably nonzero for the pilot family, without a genericity assumption. More generally suppose $a = \pi/16$, $U \ge 1$, and $U \ne b$. Put $D_4 = e^{4U} > 50$ and $\beta = \pi\cos(4a)D_4 > 100$. For $n \ge 2$,

$$\left|\frac{\phi_n(U+ia)}{\phi_1(U+ia)}\right| \le n^2\frac{2\pi n^2 D_4 + 3}{2\pi D_4 - 3}e^{-\beta(n^2-1)} \le 2n^4 e^{-100(n^2-1)}.$$

Here $\phi_1(U+ia) \ne 0$ since $|2\pi e^{4(U+ia)} - 3| \ge 2\pi D_4 - 3 > 0$. Using $n^4 \le e^{n^2-1}$ and $n^2 - 1 \ge 3 + 5(n-2)$ gives

$$\sum_{n=2}^{\infty} 2n^4 e^{-100(n^2-1)} \le \frac{2e^{-297}}{1-e^{-495}} < \tfrac{1}{2}. \tag{8.8}$$

Thus $\Phi_N(U+ia) \ne 0$ for every finite N, and $f_{N,0}(U) \ne 0$. This includes $U = 1, 3/2$, $b = 1/4$.

Theorem 8.2 (Adverse sharp-filter products on genuine stationary sets)**.** *Fix the parameters of (8.5) and assume $f_{N,0}(U) \ne 0$. For each $j = 0, 1$ there is a sequence $x_k \to \infty$ of genuine stationary points $h_{j+1}(t, x_k) = 0$ such that $h_jG_j > 0$, $h_jR_j < 0$, and*

$$\frac{-h_jR_j}{E}(t, x_k) = \exp\{(\pi/8 - a)x_k + o(x_k)\}. \tag{8.9}$$

For every fixed $0 \le \lambda < 1$, along the same sequence eventually

$$h_j(R_j + \lambda G_j) < 0, \qquad \frac{-h_j(R_j + \lambda G_j)}{E} = \exp\{(\pi/8 - a)x_k + o(x_k)\}. \tag{8.10}$$

There are also stationary sequences with $h_jR_j > 0$ and the same exponential rate for its absolute ratio. All these points eventually belong to the proved outer region, where $E > 0$ and both complete Laguerre forms are positive.

Proof. At fixed $t > 0$, the outer four-jet estimates apply for all sufficiently large x. With $A(x) = 2|Q_t(x)|$, $\omega = \log(x/(4\pi))/4$, and a continuous phase $\varphi = \arg Q_t$, their fixed-time consequences are

$$\begin{aligned} h_k &= A\omega^k\{\cos(\varphi + k\pi/2) + O(1/\log x)\}, \qquad 0 \le k \le 3, \\ \varphi'(x) &= \omega + O(1) > 0, \qquad \log A(x) = -\pi x/8 + O_t((\log x)^2). \end{aligned} \tag{8.11}$$

For $j = 0$, take phase windows centered at $\varphi = k\pi$ with half-width $\pi/6$. For all sufficiently large windows, h_1 has opposite endpoint signs and h_2 has a fixed nonzero sign throughout. There is exactly one $h_1 = 0$ point in each window, and the sign of h_0 alternates. There $|h_0| \asymp A$ and $E = h_0^2 \asymp A^2$.

For $j = 1$, center the windows at $\varphi = \pi/2 + k\pi$. Now h_2 has opposite endpoint signs and h_3 has a fixed nonzero sign. There is exactly one $h_2 = 0$ point in each window and h_1 alternates sign. At these points $|h_1| \asymp A\omega$, while $h_2 = 0$ in (8.11) gives $h_0 = O(A/\log x)$; therefore $E \asymp A^2\omega^2$.

Since $q(U) \neq 0$, the hypothesis makes each B_j a nonconstant sinusoid plus a constant. It has infinitely many fixed-width intervals on which its sign is fixed and $|B_j| \ge c_j > 0$. The spacing of the stationary points just constructed is $O(1/\log x)$. Each such interval at sufficiently large height therefore contains stationary points of both signs of h_j. By (8.6), G_j has the sign of B_j there and $|G_j| \asymp e^{-ax}/x$. In contrast, C_j is a fixed linear combination of complete physical jets, so (8.11) makes it exponentially smaller than G_j. Hence $R_j = -G_j(1 + o(1))$ on these intervals.

Choose the stationary points with $h_jG_j > 0$ or with $h_jG_j < 0$. The energy estimates above give (8.9) and its opposite-sign version: all powers of $x, \log x$ and the $O_t((\log x)^2)$ amplitude correction contribute $o(x)$ to the logarithm. Finally $h_j(R_j + \lambda G_j) = h_jC_j - (1 - \lambda)h_jG_j$, and the same comparison proves (8.10). □

Thus no constant, polynomial, or other budget $K_j(x) = \exp(o(x))$ can make the second inequality in (7.11) hold globally for this fixed sharp filter and a fixed $\lambda < 1$. This conclusion does not exclude losses locally bounded at every finite positive-time point: exponential growth in x is still locally bounded. It is a fixed-positive-time obstruction in an already favorable region, not an obstruction to an adaptive filter in the unresolved moving cusp. Neither $\lambda = 1$ nor a fraction tending exponentially fast to one is covered. In the complete product h_jC_j, the artificial endpoint terms cancel exactly.

8.4 A compact filter has no finite rational jet closure

Proposition 8.3. *For any fixed t, a, b, N, U as in (8.5), no finite family of real rational functions $r_0, \dots, r_m$ can satisfy*

$$R_j(x) = \sum_{k=0}^{m} r_k(x)h_k(t, x) \tag{8.12}$$

on a nonempty open real interval free of their poles, for $j = 0$ or 1.

Proof. Since C_j is a constant-coefficient combination of physical jets, clearing denominators would give

$$p(x)G_j(x) = \sum_{k=0}^{m} p_k(x)h_k(t, x), \qquad p \neq 0,$$

with polynomials. Both sides extend to entire functions, so this identity holds for all real x. Complete evenness and Schwarz symmetry give

$$e^{ax}h_k(t, x) = \frac{1}{2}\int_{\mathbb{R}} e^{t(u+ia)^2}\Phi(u + ia)[i(u + ia)]^k e^{ixu}\, du. \tag{8.13}$$

The Fourier density is real analytic on all of $\mathbb{R}$, and it and all its derivatives decay rapidly: holomorphy follows from the open theta strip, positive-end decay from the theta series, and negative-end decay from the complete evenness. Polynomial multiplication in x corresponds to finite derivatives in u, preserving real analyticity.

By contrast, $e^{ax}G_j$ is the Fourier transform of the compactly supported L^1 density equal to $f_{N,j}(u)/2$ on $(0,U)$, $\overline{f_{N,j}(-u)}/2$ on $(-U,0)$, and zero elsewhere. Multiplication by $p(x)$ gives a compactly supported distribution. Fourier injectivity for tempered distributions equates this compact distribution to a real analytic function. That function vanishes on (U,∞), hence everywhere. Therefore $pG_j = 0$ and $G_j = 0$.

This is impossible. The finite sum Φ_N is a nonzero holomorphic function, since every ϕ_n is positive on the positive real axis. The other factors in $f_{N,j}$ have only isolated zeros, so its compact Hermitian extension is not zero. Fourier injectivity implies G_j is not identically zero. The injectivity used here follows, for example, by Gaussian regularization and the approximate-identity limit; it requires no information about zeros of H. □

The proposition excludes rational-coefficient finite-jet identities for these fixed filters, including constant and polynomial coefficients. It does not exclude general transcendental coefficients, merely locally bounded coefficients, or moving filters. Nor can complete symmetry be restored by naively adding $\Phi_N(-z)$: as real $u \to \infty$,

$$\Phi_N(-u) = -3\pi\Big(\sum_{n=1}^{N} n^2\Big)e^{-5u} + O_N(e^{-9u}),$$

so multiplication by e^{tu^2} destroys integrability at every positive time. Complete theta evenness is an infinite-index identity. A vertical contour correction alone does not restore this raw-index symmetry. Smooth compact cutoffs still require matched endpoint derivatives and a new relative signed estimate.

8.5 Complete-source Gaussian transport and a 14-cell pilot

There is an exact alternative without sharp source endpoints. For fixed $\delta \geq 0$, multiply the complete source on the shifted contour by $e^{-\delta z^2}$, $z = u + ia$. With the same frozen a, b at both times, its retained observation and complement are

$$G_j(t,x) = C_j(t-\delta, x), \qquad R_j(t,x) = C_j(t,x) - C_j(t-\delta,x). \tag{8.14}$$

This follows directly from $e^{tz^2}e^{-\delta z^2} = e^{(t-\delta)z^2}$ and the complete contour identity. The multiplier preserves the symmetry on the imaginary segment and introduces no raw-index or upper cutoff boundary. It is generally complex on the horizontal contour:

$$e^{-\delta(u+ia)^2} = e^{\delta a^2 - \delta u^2}e^{-2i\delta au}.$$

Accordingly (7.9), which assumes real multipliers, is not applied unchanged. Formula (8.14) supplies the physical products directly. Numerical truncation errors still require full payment.

For the parameters (8.1), fix $\delta = 1/16$ before testing. The two current-time branches in (8.2) and the seven closed time slabs $[r/16, (r+1)/16]$, $1 \leq r \leq 7$, give 14 predeclared cases. All satisfy $h_jG_j > 0$, $h_jR_j > 0$, and $h_j(R_j + G_j/2) > 0$. Conservative normalized margins are

j	$Z_j\widehat{G}_j$	$Z_j\widehat{R}_j$	$Z_j(\widehat{R}_j + \widehat{G}_j/2)$
0	> 8300	> 54	> 4200
1	> 4200	> 7	> 2100

Thus this second finite pilot has favorable retained and complementary products, with the same prescribed zero loss. Stationarity is imposed only at (t, x), never at $(t - \delta, x)$.

The proof is exact rational extraction from the stored full-source certificate and requires no new quadrature. Previous and current slabs have the same spatial center and time displacements. Center coefficient differences are formed before transport. The complete spatial and temporal payments are P_{prior} for G, $P_{\text{current}} + P_{\text{prior}}$ for R, and $P_{\text{current}} + P_{\text{prior}}/2$ for $R + G/2$. All full center arrays already contain the complete index and integration tails. Intersecting with the current stationary identity $C_j = -h_{j+2} - \mu h_j$ preserves that constraint, and products of full integrated observations retain every cross term. The receipt in Appendix C records all 14 cases. It proves the stated finite comparison on $1/16 \le t \le 1/2$, not persistence of the correlation near an unknown collision or suitability of a fixed delay as $t \downarrow 0$.

The distinct real Gaussian multiplier $(u^2 - b^2)e^{-\delta u^2}$ also retains the complete source. Denote its retained residual by $G^{\text{real}}_{\delta,j}$. It satisfies

$$G^{\text{real}}_{\delta,j}(t, x) = e^{a^2\delta} C_j(t - \delta, x + 2a\delta). \tag{8.15}$$

Indeed $tz^2 - \delta u^2 = (t - \delta)z^2 + 2i\delta au - \delta a^2$. After including the physical prefactor e^{-ax}, the corresponding filtered jet is $e^{a^2\delta} h_k(t - \delta, x + 2a\delta)$. Applying $-\partial_x^2 - 2a\partial_x - \mu$ proves (8.15). Both identities converge for all real times; $0 \le \delta \le t$ is imposed when using a theorem restricted to nonnegative time. The real mask obeys the hypotheses of the paired multiplier formula, but no numerical certificate for this additional mask is asserted. These are standard Fourier/heat identities specialized to the complete theta source, not consequences of an unproved positivity property.

Their exact complementary remainders expose the next analytic obligation. Since the frozen observations satisfy $\partial_t C_j = -C_{j+2}$,

$$R_j(t, x) = -\int_0^\delta C_{j+2}(t - s, x)\, ds, \tag{8.16}$$

$$C_j(t, x) - G^{\text{real}}_{\delta,j}(t, x) = -\int_0^\delta e^{a^2 s}[C_{j+2} + 2aC_{j+1} + a^2 C_j](t - s, x + 2as)\, ds. \tag{8.17}$$

The earlier or displaced point need not be stationary.

8.6 The remaining cross-time estimate

A concrete next target is to prescribe a delay rule $0 < \delta(t, x) \le t$, a fraction $0 \le \lambda < 1$, and losses locally bounded through a candidate collision, and prove on the *current* set $h_{j+1}(t, x) = 0$ that

$$\begin{aligned} h_j(t, x) C_j(t - \delta, x) &\ge 0, \\ h_j(t, x)\{C_j(t, x) - (1 - \lambda)C_j(t - \delta, x)\} &\ge -K_j(t, x)E(t, x), \qquad j = 0, 1. \end{aligned} \tag{8.18}$$

Lemma 7.2 would then supply the required stationary loss. This is a theta-specific correlation statement: neither backward heat, complete-source symmetry, nor the transport identities prove its sign. The finite pilot establishes it only on the declared cells. Any adaptive parameters remain prescribed independently of the unknown energy; differentiating an adaptive filter additionally requires all chain-rule and moving-boundary terms. The paired source formulas and the geometric formulation identify what must be controlled, while (8.18) remains an open all-height analytic estimate.

A Complete source and integration error bounds

The following bounds make the numerical and shared-source constructions independent of an unpaid finite theta truncation. Set

$$S_k(q,d)=\sum_{r=0}^{k}\binom{k}{r}\frac{d^{k-r}r!}{q^{r+1}}=\int_0^\infty (v+d)^k e^{-qv}\,dv \qquad (q>0).$$

Let $0\le t\le\tau\le 1/2$, $0<a<\pi/8$, and $x\ge 0$. For an index cutoff N, put $m=N+1$ and

$$c_a=\cos 4a,\quad \alpha=\pi m^2,\quad \beta=c_a\alpha,\quad q_N=4\beta-9,\quad \rho_N=((m+1)/m)^4e^{-\pi c_a(2m+1)}.$$

If $\rho_N<1$, $q_N>0$, and $8\beta\ge\tau$, the complete omitted index contribution in (5.1) is at most

$$E_{N,k}=e^{-ax}\frac{(2\alpha^2+3\alpha)e^{-\beta}}{1-\rho_N}S_k(q_N,a). \tag{A.1}$$

For a cutoff $U\ge 0$, put

$$\beta_U=\pi c_a e^{4U},\quad \rho_U=16e^{-3\beta_U},\quad q_U=4\beta_U-9-2\tau U,\quad B_U=\tau U^2+9U-\beta_U.$$

If $\rho_U<1$, $q_U>0$, and $8\beta_U\ge\tau$, the entire source on $u\ge U$ contributes at most

$$E_{U,k}=e^{-ax}\frac{(2\pi^2+3\pi)e^{B_U}}{1-\rho_U}S_k(q_U,U+a). \tag{A.2}$$

Thus $E_{N,k}+E_{U,k}$ pays the omitted domain of an N-term integral on $[0,U]$. Validated retained quadrature and cell variation are additional errors.

To prove the bounds, use

$$|\phi_n(u+ia)|\le(2\alpha_n^2+3\alpha_n)e^{9u-c_a\alpha_n e^{4u}},\quad |e^{t(u+ia)^2}|\le e^{\tau u^2},\quad |u+ia|^k\le(u+a)^k.$$

For $n\ge m$, successive coefficient-majorant ratios are bounded by ρ_N uniformly for $u\ge 0$. The inequality $e^{4u}\ge 1+4u+8u^2$ gives

$$\tau u^2+9u-\beta e^{4u}\le-\beta-q_Nu-(8\beta-\tau)u^2\le-\beta-q_Nu.$$

Summing the full geometric index bound and integrating proves (A.1). For $u\ge U$, the exponent $\tau u^2+9u-\pi c_ae^{4u}$ is concave and has slope $-q_U$ at U. Its tangent bound and the geometric ratio ρ_U give (A.2). Adding the bounds overcounts their intersection, which is harmless.

For the centered residual C_j, replace any factor $S_j(q,d)$ in these payments by $S_{j+2}(q,d)+b^2S_j(q,d)$, since $|u^2-b^2|\le(u+a)^2+b^2$. For the shared carrier (5.11), set

$$\widehat{\beta}_V=\pi\cos(4a)e^{4V},\quad \widehat{q}_V=4\widehat{\beta}_V-10-2\tau V,\quad \widehat{B}_V=\tau V^2+10V-\widehat{\beta}_V.$$

If $\widehat{q}_V>0$ and $8\widehat{\beta}_V\ge\tau$, the majorant (5.12) and the same tangent argument give the full tail payment

$$e^{-ax}2(2\pi^2+3\pi)e^{\widehat{B}_V}S_j(\widehat{q}_V,V+a). \tag{A.3}$$

The same degree-two replacement pays the shared expression for C_j, because $0\le v-d_n\le v$.

For completeness, these absolute errors can be made small at the gamma exponential scale at arbitrary height. For $x\ge 41$, $p>0$, and $\ell_x=\log x$, take

$$a_x=\pi/8-1/x,\quad m=\left\lceil\sqrt{\frac{x}{2\pi}(p\ell_x+32)}\right\rceil,\quad N=m-1,\quad U=\frac{\ell_x+2\log(\ell_x+8)}{4}.$$

All preceding payment hypotheses hold, and for fixed k,

$$e^{\pi x/8}E_{N,k} = O_{p,k}(x^{5/2-p}(\log x)^{3/2}), \qquad e^{\pi x/8}E_{U,k} = O_{k,A}(x^{-A}) \quad (A > 0). \tag{A.4}$$

Indeed $c_a = \sin(4/x) \geq 2/x$, $\beta \geq p\ell_x + 32$, and $m^2 \geq 2x/\pi$. Hence $-\log \rho_N \geq 4\pi m/x - 4/m \geq 2\pi m/x$ and $(1-\rho_N)^{-1} \leq 1 + x/(2\pi m)$. Together with $e^{-a_x x} = e\, e^{-\pi x/8}$, substitution gives the first bound. For the second, $\beta_U \geq 2\pi(\ell_x+8)^2$, $U \leq \ell_x/2 + 2$, and $B_U \leq -47\ell_x^2/8 - 181\ell_x/2 - 364$. These are absolute error estimates. They neither establish a norm gap nor replace a signed estimate of the retained source. On a spatial cell, freeze the contour and cutoffs before differentiating or transporting physical jets.

B An explicit height at the condition $tL \geq 5$

The oscillatory argument in Section 2 gives the stronger asymptotic condition $tL \geq 3 + \varepsilon$, with an unevaluated height. The following elementary absolute-sum calculation supplies an explicit height for $tL \geq 5$. These are distinct quantitative statements.

Proposition B.1 (Explicit outer height)**.** *Let $L = \log(x/(4\pi))$, $\omega = L/4$, and let $Q = Q_+(x)$ be the amplitude in (2.4). If*

$$x \geq X_{\exp} := 4\pi e^{10^9}, \qquad 0 < t \leq \tfrac{1}{2}, \qquad tL \geq 5,$$

then every coordinate error in (1.9) is at most

$$\frac{108000000}{L} \leq 0.108 < \frac{1}{8}. \tag{B.1}$$

In particular, for $P_k = h_k/(2|Q|\omega^k)$,

$$P_0^2 + P_1^2 \geq \tfrac{1}{2}, \qquad P_1^2 - P_0P_2 \geq \tfrac{31}{64}, \qquad P_2^2 - P_1P_3 \geq \tfrac{31}{64}.$$

The same conclusions hold at negative heights after reflection.

B.1 Explicit weights on the complex disc

All bounds until the last sign step hold for $L \geq 1000$, $0 < t \leq 1/2$, and $tL \geq 5$. On D_x from (2.11), write $z = u + iv$ and $\theta_u = tL_u$. Then

$$u \geq x/2 \geq e^L, \qquad |L_u - L| \leq \frac{2}{xL} < 1, \qquad \theta_u \geq 5 - \frac{1}{xL} \geq \frac{19}{4}. \tag{B.2}$$

For $s = s_\pm(z)$, one has $0 \leq \Re s \leq 1$ and $|s|, |s-1| \geq u/2$. Formula (2.2) therefore yields

$$\Re\alpha(s) \geq L_u/2 - 3/u, \qquad |\alpha(s)| \leq L_u/2 + 1 \leq L, \qquad |\alpha'(s)| \leq 1/u + 6/u^2 \leq 2/u. \tag{B.3}$$

For the magnitude estimate, use $|s| \leq (u/2)\sqrt{1 + 4/u^2}$, $|\arg s| \leq \pi/2$, and the bound $3/u$ for the rational terms. It follows that

$$\sigma_\pm := \Re w_\pm \geq \frac{1}{2} - \frac{1}{2L} + \frac{\theta_u}{4} - \frac{3t}{2u} \geq \frac{3}{2}. \tag{B.4}$$

The cutoff satisfies

$$\tfrac{1}{2}e^{L_u/2} \leq N, \qquad \log N \leq L_u/2 + 1/u.$$

Indeed the upper correction is at most $\pi t/(8u)$, and flooring reduces a number at least two by at most a factor of two. Consequently, for $n \le N$,

$$e^{t\log^2 n/4} n^{-\sigma_\pm} \le n^{-p}, \qquad p = \frac{1}{2} + \frac{\theta_u}{8} - \delta, \qquad \delta = \frac{1}{2L} + \frac{7t}{4u} \le \frac{1}{32}. \tag{B.5}$$

Writing $s_0 = \theta_u - 19/4 \ge 0$, we obtain $p \ge 17/16 + s_0/8$. On the upper half-disc, the same bound holds for the first branch after multiplication by n^v: its leading real part changes from $(1+v)/2$ to $(1-v)/2$, already included in the estimate.

For $\eta > 0$ and an integer $m \ge 0$, comparison on $[n-1, n]$ gives

$$\sum_{n=2}^{\infty} (\log n)^m n^{-1-\eta} \le \int_1^{\infty} (\log y + 1)^m y^{-1-\eta}\, dy. \tag{B.6}$$

At $m = 2$, $\eta = 1/16$, the right side is $2/\eta^3 + 2/\eta^2 + 1/\eta = 8720 < 9000$. Hence

$$\begin{aligned} |S_{t,N}(w_\pm) - \zeta(w_\pm)| &\le \frac{9000}{4L_u}(19/4 + s_0)2^{-s_0/8} + 2N^{-1/2} \\ &\le \frac{29250}{L_u} + 3e^{-L_u/4} \le \frac{31000}{L}. \end{aligned} \tag{B.7}$$

Here $s_0 e^{-s_0 \log 2/8} \le 8/(e\log 2) < 8$, so the time-dependent numerator is less than 13. Moreover, $29250/(L-1) < 30000/L$ for $L \ge 1000$, while $e^{(L-1)/4} \ge (L-1)^2/32 \ge 3L$. Thus the displayed bound includes both the weighted discrepancy and the complete ordinary zeta tail, uniformly even as tL grows.

B.2 All imported approximation errors

On the upper half-disc, (2.6) gives $|\gamma| \le 2$ and

$$|\kappa| \log N \le \frac{tv}{2(u-6)}(L_u/2 + 1/u) < \tfrac{1}{2}, \qquad N^{|\kappa|} \le 2.$$

Since $\zeta(17/16) \le 17$, the full positive weight sum in (2.7) is at most $5 \cdot 17 = 85$. Denote that equation's exponent by

$$q_n = \frac{(t^2/16)\log^2(u/(4\pi n^2)) + 0.626}{u - 6.66}.$$

The estimates $-2/u \le \log(u/(4\pi n^2)) \le L_u$, $(L+1)^2/64 + 1 \le L^2/32$, and $u - 6.66 \ge u/2$ imply $0 \le q_n \le L^2/(16u) < 1$. Using $e^q - 1 \le 2q$ on $[0,1]$ gives

$$e_A + e_B \le \frac{85L^2}{8u} \le \frac{11L^2}{u} \le \frac{1}{L}. \tag{B.8}$$

For the final inequality, $u \ge e^L \ge L^4/24 \ge 11L^3$ suffices.

The positive correction exponent in (2.8) is at most

$$\frac{1.24(3^v + 3^{-v})}{N - 0.125} + \frac{3|L_u + i\pi/2| + 10.44}{u - 12} \le 20e^{-L_u/2} + 7L/u \le 1.$$

To verify the last inequality, use $20e^{-(L-1)/2} \le 160/(L-1)^2 < 1/2$ and $7Le^{-L} \le 168/L^3 < 1/2$. Its negative main exponent therefore gives

$$e_{C,0} \le e\exp\{-(1/4 + 19/64)L_u\} \le 3e^{-L_u/2} \le \frac{1}{L}, \qquad e_A + e_B + e_{C,0} \le \frac{2}{L}. \tag{B.9}$$

All three errors in the imported theorem are included; the constant 0.626 in q_n is independent of t.

B.3 Amplitude comparison and all four derivatives

Let $\Gamma = |M_t(s_+(x))| = |M_t(s_-(x))|$. On D_x,

$$\left|\frac{d}{dz}\log M_t(s_\pm(z))\right| \le \frac{|\alpha|}{2}(1+t|\alpha'|/2) \le L.$$

Integration along a segment of length at most $1/L$ yields $|M_t(s_\pm(z))| \le e\Gamma < 3\Gamma$. Also (B.4) and the zeta and reciprocal-zeta Dirichlet series give $|\zeta(w)|, |1/\zeta(w)| \le \zeta(3/2) \le 3$, hence $\Gamma \le 3|Q|$. Combining (B.7) and (B.9) with (2.5) gives

$$\begin{aligned}|H(t,z) - \mathcal{P}_t(z)| &\le 2(3\Gamma)\frac{31000}{L} + (3\Gamma)\frac{2}{L} = \frac{186006\Gamma}{L}\\ &\le \frac{558018|Q|}{L} \le \frac{560000|Q|}{L}.\end{aligned} \tag{B.10}$$

Schwarz symmetry extends this upper-half-disc estimate to the full disc. Cauchy's inequality now gives

$$\frac{|\partial_x^k(H(t,\cdot) - \mathcal{P}_t)(x)|}{2|Q|\omega^k} \le \frac{280000\,k!\,4^k}{L} \le \frac{107520000}{L}, \qquad 0 \le k \le 3. \tag{B.11}$$

For the analytic carrier, at $s = (1+ix)/2$ direct differentiation of (2.2) gives

$$|\alpha - (L/2 + i\pi/4)| \le 4/x, \quad |\alpha'| \le 2/x, \quad |\alpha''| \le 3/x^2, \quad |\alpha'''| \le 9/x^3. \tag{B.12}$$

The first estimate follows from $s = (ix/2)(1 - i/x)$ and $|\operatorname{Log}(1 - i/x)| \le 2/x$. For the others one uses

$$\alpha' = -\frac{1}{2s^2} - \frac{1}{(s-1)^2} + \frac{1}{2s}, \qquad \alpha'' = \frac{1}{s^3} + \frac{2}{(s-1)^3} - \frac{1}{2s^2}, \qquad \alpha''' = -\frac{3}{s^4} - \frac{6}{(s-1)^4} + \frac{1}{s^3}.$$

Let $J(x) = \log M_t(s(x))$. Then

$$\begin{aligned}J' &= (i/2)\alpha(1 + t\alpha'/2),\\ J'' &= -\tfrac{1}{4}\{\alpha' + (t/2)[(\alpha')^2 + \alpha\alpha'']\},\\ J''' &= -\tfrac{i}{8}\{\alpha'' + (t/2)[3\alpha'\alpha'' + \alpha\alpha''']\}.\end{aligned}$$

Equations (B.3) and (B.12) imply $|J' - i\omega| \le 1$, $|J''|, |J'''| \le 1$ and $|w'|, |w''|, |w'''| \le 1$. A majorant for the first bound is $\pi/8 + 2/x + L/(4x) < 1$; the remaining bounds follow by inserting the stated powers of x^{-1} and $x \ge e^L$.

The prime-power series and (B.6) at $\eta = 1/2$ give

$$|(\log\zeta)'(w)| \le 6, \qquad |(\log\zeta)''(w)| \le 26, \qquad |(\log\zeta)'''(w)| \le 158.$$

Thus for a local logarithm $g = \log Q_+$, the chain rule yields

$$|g' - i\omega| \le 7, \qquad |g''| \le 1 + 26 + 6 = 33, \qquad |g'''| \le 1 + 158 + 3\cdot 26 + 6 = 243.$$

The resulting carrier errors are

k	$\dfrac{\lvert Q_+^{(k)}/Q - (i\omega)^k\rvert}{\omega^k}$
1	$28/L$
2	$56/L + 1312/L^2$
3	$84/L + 3936/L^2 + 81856/L^3$

(B.13)

Every entry in the right-hand column is at most $100/L$ for $L \geq 1000$. At order three, for instance, expanding $(i\omega + d)^3 + 3(i\omega + d)g'' + g'''$ with $|d| \leq 7$ bounds the unnormalized discrepancy by $21\omega^2 + 246\omega + 1279$. The conjugate branch contributes the same normalized bound; the factor two is already included in $2|Q|\omega^k$. The carrier error at order zero is exactly zero.

Adding (B.11) and (B.13), every coordinate error is at most $107520100/L < 108000000/L$. For $L \geq 10^9$ this is less than $1/8$, and (2.31) proves Proposition B.1.

The value X_{exp} is an explicit consequence of the primary approximation and elementary inequalities. Its exponential need not be evaluated numerically. The accompanying exact checker verifies 52 rational budgets and sufficient polynomial inequalities for every $L \geq 1000$ by shifting $L = 1000 + y$ and checking nonnegative coefficients. It audits the arithmetic in this proof, rather than replacing its analytic arguments. The bound supplies no practical finite-strip coverage between 104 and X_{exp}, and it does not evaluate the height in the stronger threshold-three theorem.

C Reproducibility and scope of the checks

All paths in this appendix are relative to `papers/investigations-2/active/21_quaternion_theta_geometry/` in the accompanying repository. The analytic proofs above are distinct from the finite interval decisions. The latter use complete error bounds, outward rounding, and closed-cell coverage; hashes identify the evaluated sources and records, but do not substitute for their mathematical content.

The bounded results can be regenerated using the drivers `numerics/certify_compensated_stationary_time_extension.py`, `numerics/certify_correlated_source_height_extension.py`, and `numerics/certify_phase_centered_residual.py`. The correlated-height guide `numerics/CORRELATED_SOURCE_HEIGHT_README_20261014.md` gives the parameters for its three bands. The full-time certificate is `numerics/COMPENSATED_STATIONARY_TIME_EXTENSION_20261014.json`; the final centered-band record is `numerics/THETA_CENTERED_RESIDUAL_100_104_20261014.json`. The independent internal replays are documented in Reviews 6, 9, and 11 in `reviews/`. They reproduced the numerical records after removal of runtime metadata.

For compact exact audits, run the following from `numerics/`:

```
python3 audit_correlated_source_height_records.py
python3 audit_theta_centered_residual_record.py
python3 check_octonion_jet_identities.py
python3 check_theta_centered_residual_identities.py
```

The centered-band auditor additionally requires `-record THETA_CENTERED_RESIDUAL_100_104_20261014.json`. Each requires the option `-output` followed by a desired JSON path. The first two inspect the retained records and bound source versions; the latter two check 248 and nine exact algebraic identities, respectively. The octonionic identities, centered residual, and outer analytic argument have separate internal assessments in Reviews 10–12. The diagnostic `numerics/check_outer_cusp_and_phase_obstructions.py`, also run with `-output`, requires python-flint and imports `numerics/check_shifted_contour_payments.py`. Its retained record `numerics/OUTER_CUSP_AND_PHASE_OBSTRUCTION_CHECKS_20261014.json` contains the pointwise counterexamples in Section 7, twelve staircase boundary checks, exact four-jet perturbation algebra, and selected complete-source point enclosures. It does not certify a new rectangle or prove the uniform asymptotic theorem.

The subsequent paired algebra, explicit outer constants, and signed-split controls are audited by `numerics/check_signed_carrier_pair_identities.py`, `numerics/check_explicit_outer_constants.py`, and `numerics/check_signed_curvature_split.py`. They require only the Python standard library and an `-output` path, and check 34, 52, and 14 exact statements respectively. Their records are `numerics/SIGNED_CARRIER_PAIR_IDENTITY_CHECKS_20261014.json`, `numerics/EXPLICIT_OUTER_CONSTANT_CHECKS_20261014.json`, and `numerics/SIGNED_CURVATURE_SPLIT_IDENTITY_CHECKS_20261014.json`. These checks support algebra and explicit inequalities; the infinite source and asymptotic arguments require the proofs in the text.

The full-residual branch comparison is recorded in `numerics/STATIONARY_FAVORABLE_CURVATURE_PILOT_20261014.json`. The proper source-filter and Gaussian pilots are produced by `numerics/certify_predetermined_source_filter_pilot.py` and `numerics/check_gaussian_source_filter_pilot.py`. Each accepts `-record`, `-original-driver`, `-contour-source`, and `-output`; the inputs are the centered-band record, its driver, and `numerics/check_shifted_contour_payments.py`. The proper-filter run uses python-flint 0.9.0 at 192 bits and retains all 64 classifications, including 32 failed absorption conditions. The Gaussian extraction uses exact rational arithmetic and retains all 14 cases. Their records are `numerics/PREDETERMINED_SOURCE_FILTER_PILOT_20261014.json` and `numerics/GAUSSIAN_SOURCE_FILTER_PILOT_20261014.json`. Reviews 14–16 document the analytic audits and independent replays, including the eight full-residual branch slabs. Review 17 audits the present manuscript integration; its fresh exact checks are distinct from the preceding full interval replays.

The main remaining obligation is the one-sided inner-region bound (7.5), with all cross terms retained and losses locally bounded through candidate positive-time collisions. The complete-source Gaussian identities make a specific cross-time correlation available for that task. Improving the explicit height and connecting the finite cover remain separate quantitative work. Neither a fixed-time asymptotic nor a ratio divided by an unknown physical energy supplies the missing bound.

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